

II

A Better Picture of Social Change

Hidden Histories of Social Change: People Who Change Systems That Change People

It's Always a Duck and It's Always Also a Rabbit

One of the world's most famous visual illusions was first published in the German humor magazine *Fliegende Blätter* in 1892. It will be familiar to anyone who has taken an intro psychology class (figure 4.1). Some people see a duck facing to the left, while others see a rabbit looking to the right. Most people assume there's only one way to see it until something prompts a second look. The illustration serves as a reminder that the very same thing can be understood in completely different ways.

Researchers showed this image to hundreds of people, from ages two to ninety-three, entering the Zürich Zoo in Switzerland. When they conducted the exercise on a random Sunday in October, most zoo-going Zürichers saw a bird of some kind, like a duck or a stork. When they conducted it on Easter Sunday, though, a large majority (including 90 percent of young children) saw a bunny.

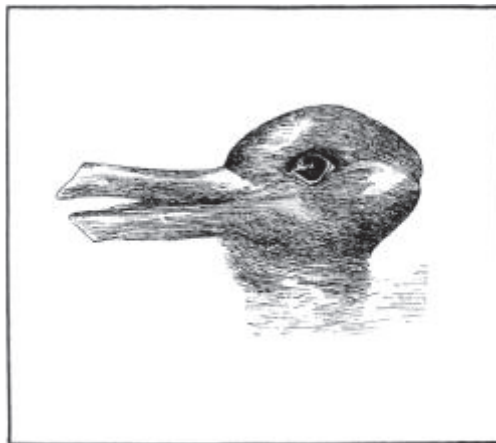


Figure 4.1

The duck-rabbit.

“But when you get down to it, taking away the influence of Easter or whatever,” someone might ask, “what is it *really*—a duck or a rabbit?” The answer is that it’s both. The image is not really a duck. And it’s not really a rabbit. It’s a duck-rabbit. A single picture can be interpreted in more than one way. So when we (Michael, Alex, and Dan) see a duck-rabbit, we see a reminder to reject false choices. We see a reminder to question the way a problem, or a history, or a program for changing the world, is framed.

Social events can also be interpreted in different ways. Depending on how they frame it, one storyteller can make an event look like a vindication of the importance of individual action. Another storyteller can make the very same event look like a demonstration of the priority of structural change. Consider the following:

E-scooters: Italy awards 500€ grants to city dwellers who purchase bicycles or e-scooters. These awards can be framed as an individualist reform because they aim to encourage citizens to reduce their carbon footprint by reducing their personal use of automobiles. They can also be framed as a structural reform because they’re made possible by a far-reaching nationwide public policy that aims to change the financial incentives that structure individuals’ choices.

Phone banking: You volunteer with an organization to make phone calls to persuade voters in your town to reform the racist and classist cash bail system. This effort can be framed as individualist because you're trying to persuade other people to change their votes one by one. This action can also be framed as structural because it is part of a collective action movement seeking to change laws and ultimately to change the structural conditions for poor people who end up entangled with the criminal justice system.

COVID-19: The economic slowdown caused by the coronavirus pandemic [reduced global carbon pollution](#) by over 6 percent in 2020 as compared to 2019. This was the largest single-year drop in modern history. The temporary reduction in emissions was framed by some as vindicating the power of individual action, because it showed the enormous changes people can make if they choose to alter their behavior. It was framed by others as vindicating the importance of larger structural forces because the individual behavioral changes resulted from a profound "shock to the system" that was enforced by emergency, top-down, state-based policy.

Which lesson someone comes away with depends a lot on how the story's told. One journalist pitches the pandemic-based drop in pollution as a story about the power of individual behavior change, while another highlights the need for structural reform. We ask, [Why not both?](#)

Each example above involves individuals who help initiate change. Somebody in Italy has the idea for e-scooters. Somebody in the activist organization sets up the phone bank. Each example also requires individual people to implement the change after it's set in motion. Somebody needs to advertise the e-scooter rebates; somebody has to cut the checks. And each change will wither away if people don't persist in their efforts to sustain it: Italians need to buy the scooters, ride them, and pester their friends and neighbors to do the same. Phone bankers have to make call after call after call. Motivated individuals are crucial to initiating, implementing, and sustaining change.

But none of this happens in a vacuum. Each individual action occurs within a set of social structures, a broader cultural and physical landscape: roads to ride the scooters on, phone networks, neighbors willing to take calls. Individual actions are predicated on structures like these. What's more, the individuals in these cases are trying to make changes to the structures in which they operate. Some aim to spread new norms for a low-carbon lifestyle, others to end cash bail. These structures are not easy to keep in view. Individualist cultures like the US often relegate social structures to the unnoticed background, like the duck on Easter Sunday. Instead, most Americans are almost inevitably drawn to individual people, much as one's eyes are drawn to the figure in the center of a photo.

Resisting false choices is an important step forward in seeing ourselves in structural change. Before the rest of this book makes the case for a both/and approach to repairing contemporary injustices, this chapter explores two historical case studies. In each, we tease out overlooked features that are hard to notice when too much attention is given to individuals in isolation or to social structures in isolation.

Getting MADD

On May 1, 1980, Clarence William Busch was arrested for drunk driving, his fifth DUI in four years. His car was impounded, but he was released on bail the next day and allowed to keep his driver's license. After taking his wife's car to a bar, Busch, drunk again, veered into a bike lane and killed thirteen-year-old Cari Lightner.

This was the worst, but not the first, of the Lightner family's horrendous encounters with reckless drivers. A drunk driver had crashed into and injured Cari's grandmother and sister Serena, and her four-year-old brother Travis spent four days in a coma after being hit by a driver impaired by tranquilizers. With Cari's death, her mother Candace had had enough, and four days after the funeral [she started MADD](#).

MADD stands for "Mothers Against Drunk Driving," and it's a remarkable organization. A 2005 Gallup poll found that an amazing 94 percent of Americans were familiar with it. MADD is an exemplar of the many citizen activist groups that flourished in the 1970s and '80s, and it may be one of the most

successful. From 1980 to 2004, [alcohol-related traffic deaths](#) decreased from an estimated 30,000 per year to 16,694. For individuals under twenty-one, these deaths had decreased by 83 percent by 2021.

Nobody knows for sure how instrumental MADD was in creating these changes, but there's reason to think the group was influential. Media coverage of alcohol-impaired driving shot up in the '80s, often featuring representatives from MADD. The group lobbied state legislatures to increase the penalties for impaired driving, and MADD leaders were typically present at the signing of these bills. Candace Lightner was involved in, and present at, the signing of the National Minimum Drinking Age Act of 1984, which compelled states to raise the drinking age to twenty-one. Good evidence suggests that enacting this law was a structural change that [effectively reduced drunk driving](#).

One way to tell this story focuses on Lightner herself, the charismatic individual who inspired many more individuals, starting a crusade that went on to make a big difference. Her efforts led to a shift in popular attitudes about drunk driving, changing the way people think, feel, and act. In addition to the official laws, social norms around drunk driving have also transformed since the 1970s, in part because of Lightner and MADD's efforts. It's now widely understood that [driving drunk is shameful](#) and friends and family, bartenders, and bystanders share the responsibility to intervene and prevent it. A full accounting of the decline of drunk driving would have to note how one individual sparked a movement, inspiring other individuals to change a whole lot more individual hearts and minds.

These successful efforts to change widespread perceptions and norms around drunk driving were in turn integral to getting all those bills passed. Building a new consensus helped Lightner and other activists persuade individual policymakers to introduce pivotal legislation. It made it easy for legislators who had been on the fence to vote in favor of the new bills without hurting their chances to get reelected. It also helped motivate agents of the state to carry out all the new rules and regulations.

Newly passed and enforced laws then fed back into and reinforced evolving perceptions and norms around drunk driving. All this gradually coalesced into a new facet of the background structure of American life. This is precisely the pathway to change that Margaret Mead had in mind when she reportedly said, "Never doubt that a small group of thoughtful, committed citizens can change the world. Indeed, it's [the only thing that ever has](#)."

But there is always more to the story. How, for example, was Lightner able to persuade so many people? To understand how her message broke through, it can't be examined in isolation. The broader culture and network of social structures in which Lightner was operating must be taken into account, as they provided the conditions for her message to take root and thrive.

Mothers Against Drunk Driving wasn't the group's original name. When Lightner started the group, it was ["Mothers Against Drunk Drivers."](#) Lightner herself didn't think the fundamental problem was the drinking age or that the solution was Big Structural Change. She unrepentantly blamed drunk driving on reckless individuals. "This was not an 'unfortunate accident,'" she told *People* in 1981, "Cari was the victim of a violent crime. If my daughter had been raped or murdered, no one would say of the killer, 'There but for the grace of God go I.' Death caused by drunk drivers is the only socially acceptable form of homicide."

It's likely, then, that part of MADD's success derived from the ways Lightner tapped into the shared culture of individualism we've explored in previous chapters. Her emphasis on individually irresponsible drivers resonated with the general law-and-order stance of a then very popular President Reagan, who endorsed the organization. While the Crying Indian ad said that people start pollution and people can stop it, Lightner was saying that (drunk) people cause accidents, and (sober) people can stop them. The case of Candace Lightner getting MADD shows how movements gain a foothold when their messages harmonize with their cultural moment.

While the Ad Council and Keep America Beautiful invoked personal responsibility to block structural change, Lightner's appeal to personal responsibility brought structural change. The American culture of individualism itself formed a key element of the structural background that empowered her committed

group of individuals to bring about changes to norms and laws. By playing to the ambient individualism humming in [the background of American minds](#), Lightner was able to get her hands on the levers of power.

Even the campaigns created by the Ad Council and Keep America Beautiful—the campaigns rightly criticized for placing the blame for litter on individuals rather than companies—did some good. It's nice to live in a country without roadside litter, which declined dramatically in the '70s. More important, it's difficult to imagine the creation of landmark laws like the Endangered Species Act and the Clean Air Act outside the context of the early days of the environmental movement, even with its worrying focus on personal responsibility. The United States has some of the [cleanest air in the world](#) today.

Of course, there are plenty of ways to promote structural change without demonizing individual people the way Lightner did. Hers is not the only roadmap to social change. Rather than using persuasive strategies that exploit a prevalent culture of individualism, many social movements in the US and abroad have sought to resist that very culture. One way people have tried to do this is by building—literally, physically building—new spaces for communal culture to flourish.

Go Down, Moses

Modern-day Copenhagen is a haven for pedestrians and cyclists (figure 4.2). Each day in the city, half of all commutes to school and work are taken by bicycle. [Copenhageners are happy with it](#) too, with 97 percent saying they're satisfied with city cycling conditions. But Copenhagen wasn't always like this. Figure 4.3 shows the same square in 1960; figure 4.4 shows similar pictures from [Amsterdam in 1971 and 2020](#).

What happened? European cities like Copenhagen and Amsterdam faced pressure in the 1960s to modernize. Big US cities, with their freeways, wide roads, and suburbs, were seen as the future. In Great Britain, large parts of Birmingham were demolished to build a ring road, turning footpaths into underpasses. In 1966, Stockholm city planners sought to connect the islands of the city with the six-lane Essingeleden highway. It remains an auto-dense city. Copenhagen was headed the same way. In 1958, the city proposed a twelve-lane highway that would cut across the city center and pave over parts of its urban lakes, with plans to [raze the Vesterbro neighborhood](#) for motorways.



Figure 4.2
Copenhagen, 2018.



Figure 4.3
Copenhagen, 1960.



Figure 4.4

Amsterdam, 1971 and 2020.

Fortunately, a small but developing group of urban planners in the “livable cities” movement agitated for a different approach and advocated organizing the city around bikes and buses. [Shopkeepers pushed back](#), arguing they’d lose customers without car traffic. It wasn’t obvious which way things would go until an unexpected shock to the system: the 1973 global oil crisis. Gas became sharply more expensive, car culture suddenly looked much less appealing, and cycling took off. It’s been growing ever since.

Overhauling major cities in this way may sound a bit like magic to people living in places with dysfunctional politics. But similar triumphs have happened elsewhere. For example, Washington Square Park in Manhattan [used to be a parking lot \(figure 4.5\)](#). It was also nearly turned into a highway. The indomitable Robert Moses wanted to remove the park’s fountain and run what he dubbed the Lower Manhattan Expressway right through the middle. At the time, Moses seemed to have the power to do anything. Despite having never been elected to office, he held [twelve official public titles](#) simultaneously. He also controlled construction for all of New York City’s Housing Authority, built the Triborough Bridge (now the Robert F. Kennedy Bridge), most of the city’s pools, and basically created suburban Long Island. Robert Caro’s Pulitzer Prize–winning biography *The Power Broker* is a 1,200-page display of Moses’s [influence, ego, and racism](#). Caro recalls watching Moses hang up his office phone and say, “They expect me to build playgrounds for that scum floating up from Puerto Rico.”



Figure 4.5

Washington Square Park, Manhattan, circa 1960–1969. Image via NYU Local.

But Moses failed to pave over Washington Square Park. He met his match in one Jane Isabel Butzner from Scranton, Pennsylvania. By that point, Jane Butzner had become Jane Jacobs, journalist for *Architectural Forum* magazine. A social change entrepreneur, Jacobs would end up writing ten books about city life, inequality, and self-government, including *The Death and Life of Great American Cities* (1961), an influential critique of urban renewal programs. At the time, “urban renewal” was one of those code words like “redlining,” a bland euphemism for razing Black and brown neighborhoods to build more highways, pools, and parks, predominantly for white people.

In opposition to Moses, Jacobs formed the catchily named Joint Emergency Committee to Close Washington Square to Traffic. The name was her idea, and she took pride in it. “Always put what you are trying to achieve in the name of the committee,” a friend recalls Jacobs saying, “because most [people will read no further](#) than that.” Jacobs pieced together the labyrinthine relationships between New York City agencies and managed to halt Moses’s plans. Her efforts culminated with a successful petition of the “Board of Estimate” to direct the traffic commissioner to create a three-month trial of closing the park to traffic. The trial was a hit, too popular to reverse. On June 25, 1958, the city closed Washington Square Park to traffic for good. They held a ribbon *tying* ceremony to celebrate.

One lesson from these histories is that good urban planning matters. The built structure of a city can make the difference between a pedestrian utopia and gridlocked traffic-choked streets. Two waterfront urban hubs as culturally and historically similar as Stockholm and Copenhagen can be worlds apart when it comes to city cycling. (That said, both cities have [extensive public transportation networks](#) of subways and buses, which makes them both worlds more livable and accessible and less polluted than, say, Los Angeles.) The right kinds of urban design and built environments, in turn, promote a stronger sense of community and discourage a culture of individualist isolation. No wonder that, by some measures, Denmark has the [highest levels of social trust](#), fellow feeling, and happiness in the world, whereas the US is at best in the middle of the pack.

Given the power of urban planning to shape the quality and community of city life, score one for structural reform. But also note that committed individuals had to put these structures in place and fight to preserve the ones they already had. You might have paid for parking in Washington Square Park on your last trip to the Big Apple if it hadn’t been for Jane Jacobs.

From Either/Or to Both/And

Suppose you too are passionate about making cities and neighborhoods more livable, safe, and healthy. You've recently visited Copenhagen and you're fired up to promote bike commuting in your hometown. You talk to some friends—maybe older, liberal friends—and they tell you:

Change begins within. Focus on your own choices and what you can control. Start riding a bike to work. Join a cycling club! Never doubt that a small group of thoughtful committed individuals can change the world! Vote!

But then you talk to some other friends—some younger, more radical friends—and they tell you the opposite:

Ride your bike if you feel like it, but don't mistake it for activism. What you do on your bike is just a drop in the bucket. It's dangerous without protected bike lanes anyway. City planning is a structural problem, and structural problems require structural solutions. Don't focus on yourself, change the system!

This kind of debate between your two groups of friends succumbs to the either/or error. It's also an instance of what philosophers call a [“category mistake,”](#) which is just a fancy way of saying your friends are comparing apples to oranges. If you, an isolated person, decide to ride your bike to work every day, chances are this will have a small impact on the course of human history. If a city government—a legal entity empowered to build roads, enforce laws, and determine where people live—decides to reduce street parking in favor of promoting bike lanes, this will likely have a bigger impact. If all the nations of the world come together to sign an enforceable treaty to plaster every country with bike lanes, then the impact, presumably, would be bigger still. But these are comparisons between a small thing an individual can do, a bigger thing a city government can do, and an even bigger thing that nations can do together.

We could just as well draw the contrast differently. If you, a single person, decide that biking to work is not enough and instead dedicate your life to criticizing “urban renewal” projects, and you write ten books about it, study the unnoticed levers of power in your city, and enlist influential friends to your cause, then you, an individual, will have a better chance—like Jane Jacobs—of making a bigger impact. Meanwhile, if a government is feckless and stymied by infighting or partisanship, or staggers from bankruptcy to bankruptcy, chances are that its “big,” “structural” urban planning will have a small impact. The best-laid plans of social institutions often go awry (or go nowhere).

More useful comparisons avoid these category mistakes. Rather than engaging in an unanswerably abstract debate about whether individual change matters at all, someone could compare one specific thing an individual might do to another specific thing that individual might do. A person moving to a new place might [consider the many pros and cons](#) of living close to work and commuting by bike, or living further away and commuting by bus, or living even further away and taking the money saved on housing to buy an electric car.

Meanwhile, people can compare, say, one urban plan to another. Given limited financial resources, should a town trying to go green focus on building more sidewalks, or more bike lanes, or more public transportation, or more electric vehicle (EV) charging stations? Or perhaps the town can drum up the resources to pursue all these by hiking taxes on its wealthiest residents and cutting spending elsewhere, say, by skipping next year's purchase of militarized police gear?

These kinds of comparisons—between tangible individual changes or between tangible structural changes—avoid pitting apples against oranges. We have to start with comparisons that make sense. Once

we do that, we can start bringing more nuance on board. Comparing tangible things individuals can do requires paying attention to the structural background, and comparing tangible structural changes requires paying attention to the individuals in the foreground.

For example, the promise of any policy proposal for a town deliberating about how best to go green will depend on what individual citizens think about the policy. People with specific values and viewpoints—design planners, engineers, voters, government officials, activists, lobbyists—will have to initiate, implement, and sustain the change it specifies. All these individuals have a say in which structural reforms, if any, get made.

And for every structural change to promote, there are actions that individuals should take to promote that structural change. Which actions any particular person might take will depend on which roles they occupy (urban designer, parent, voter) and how much bandwidth, money, and time they have to devote to the cause. If the community settles on more bike lanes, then individuals who support that decision could, for example, start buying and riding bikes, inviting their friends on Sunday morning group rides, handing out pro-bike lane flyers, opening up a used bike shop, and so on.

The vice versa is equally vital: for every individual behavior change to pursue, there will be structural changes to put in place because they enable individuals to make those changes. What kinds of structural changes will get people to buy and ride bikes? Cash incentives would probably help (see Italy's rebates for e-scooters). Maybe the town, or a group of volunteers, starts offering free cycling lessons or local tours for children and adults. Maybe teachers at the school redesign their curricula to ensure that kids are learning about the virtues of bike lanes in class. [Schools can start "bike buses,"](#) morning group rides making it safer for little kids to pedal to class. Maybe residents donate money to the school to start an afternoon cycling club for students. Maybe they take field trips to go mountain biking or watch a race.

The mistake is to compare an individual change to a structural reform (one person taking up cycling versus a whole town creating bike lanes). The solution is to compare various individual changes to each other, some of which have better potential to scale up to structural change, and to compare various structural reforms to each other, some of which have better potential to be implemented and sustained by individual people.

Another way to put this is to think in terms of bundles. What is the bundle of individual changes that people can make to promote the right kinds of structural changes that will in turn enable individuals to make the most impact? The best bundles will be those that are sensitive to the culture of the communities they are designed for. They'll aim [to create feedback loops](#) between individual and structural reforms.

The Real Remaining Trade-Offs: Effort and Attention

When we say there's no sense to be made of choosing between changing individuals or changing structures, it might sound like we're denying that there's any meaningful distinction between individuals and social structures at all. We're not. There is a difference between them, just like there is a difference between figure and background in a photo. Both are real, both are present, and both are essential. That said, people can't focus on everything everywhere all at once. There are only so many hours in a day, and everyone has to set priorities. They have to choose what to pay attention to, and that might mean sometimes paying relatively *more* attention to individuals or to structures.

In some cases, people might be able to focus on how best to design social systems without worrying too much about the details of individual psychology. Say a policymaker wants to make it as easy as possible for people [to show up for court](#). They don't need a doctorate in neuroscience to predict that offering people multiple attendance options (e.g., online via Zoom or in person, morning or night, weekday or weekend) will boost attendance rates.

Other times, it's important to let the structures remain in the background and keep specific individuals front and center. Say someone hears their white friend ask about touching a Black stranger's hair. They don't need a PhD in ethics to know this isn't OK.

Examples like these show the importance of being able to shift focus between individuals and structures as the situation demands. Seeing when to focus on one or the other requires understanding how they're related, just like a songwriter knows when the melody or rhythm of a song needs to be changed by appreciating how they're related.

People can't get policymakers to sign off on allowing multiple court attendance options without persuading them to do it. Likewise, people can't understand everything wrong with asking to touch Black people's hair without knowing the history of racist Europeans [treating Black bodies as exotic](#) oddities to put "on display." The full meaning of an individual action depends on the history and structures shaping it, and the possibility of structural change depends on the individuals creating and sustaining it.

We next turn from the past to the present and future. We'll offer a novel take on what many hold up as the Most Distractingly Individualist Approach of them all: diversity trainings.

Unlearning the Habit of Racism

The Racialization of American Politics

President Barack Obama took a risk in 2009 when he decided to make healthcare reform the top priority of his first term. He had to choose between several deserving ways to spend his political capital. Climate change? Immigration reform? His healthcare initiative also followed in the wake of recent Democratic failures. Hillary Clinton's efforts to reform medical insurance in 1993 were still remembered as a political debacle. The system remained untouched, and Democratic strategists came to think healthcare was a political minefield. Insurance companies were too powerful and Americans too attached to their doctors to risk attempting an overhaul of the system.

But [Obama was taking another risk](#), less obvious at the time. He knew well that American politics had become increasingly polarized, with "blue states" and "red states" far apart on many issues, including healthcare. What Obama did not see coming was just how much American politics had become racialized. As it turned out, he was betting the success of healthcare reform on navigating the tricky waters of American attitudes about race.

This story and others like it foretell that the prospects of overhauling entrenched American systems hinge on the prospects of uprooting entrenched American prejudices. One lesson from chapter 2 seemed to be that common efforts to reduce prejudice—through diversity training and antibias workshops—are ineffective. They're often individualist distractions from deeper structural problems. But if there's no way to reform big structures like healthcare without reducing prejudice *and* there's no effective way to reduce prejudice, does that mean we're doomed? Fortunately not. Under the right conditions, real prejudice reduction is possible. And prejudice reduction is necessary.

Everything Is Two Degrees from Race

The political scientist Michael Tesler studies how people's attitudes about race shape their views about politics, culture, and public policies. Through a series of studies, Tesler showed that white Americans' thoughts about Obama's Affordable Care Act (ACA) were predicted by their racial attitudes. As Obama himself became the face of healthcare reform—even embracing the branding of it as "Obamacare"—the influence of those racial attitudes became stronger and stronger. The racial divide between Black and white Americans' support for healthcare reform [ballooned by 20 percentage points](#) between Clinton's plan in 1993 and Obama's in 2009.

Why? [Tesler had a hunch](#), as he said in one interview, that race was "simply more on the top of people's heads when it comes to Barack Obama." In 2009 the president was asked on *Meet the Press* about the possibility, expressed by former President Jimmy Carter and members of the Congressional Black Caucus, that racism was behind the growing hostility toward the ACA. Obama acknowledged that race could be a factor: "Sometimes they vote for me for that reason, sometimes they vote against me for that reason." But he tried to assure anxious voters that "this debate that's taking place is not about race."

Try as he might, Obama failed to keep race out of people's minds when it came to healthcare reform. Tesler and his colleagues looked closer and found a causal connection between racial resentment and people's support for the ACA. In one experiment, they described the details of the ACA to one group of voters as part of Obama's proposed reforms, to another group as part of President Clinton's 1993 reform efforts, and to a third group as simply being proposed by "some people." [The most racially resentful white voters](#) opposed healthcare reform in all three cases, but they were almost twice as likely to dislike elements of the ACA when they associated them with Obama instead of Clinton.

The racialization of American politics isn't new, but its strength and scope have grown dramatically since the 1980s. In some cases, the link between people's racial attitudes and their political beliefs is unsurprising. Take views about welfare and crime. It's relatively easy to see how pernicious racial stereotypes about certain people being more prone to violence or less willing to "try hard" could increase support for harsher crime and welfare policies. It's more surprising that racial attitudes inform people's beliefs about things like Social Security, tax policy, and the Iraq War. According to some scholars, race has "eclipsed class as the organizing principle of American politics" since the 1960s.

There's a similar story to tell about race and climate change. Most people know that Republicans tend to be less concerned about climate change than Democrats. What most people don't know is how big a role race plays in this disparity. Among the least racially resentful white Republicans, only 7 percent deny that climate change is happening. Among the most racially resentful, the percentage jumps to 26. What about the reality that climate change is caused by humans? A full 84 percent of white Republicans high in racial resentment do not believe that human activities are driving climate change. Put it this way: if you wanted to guess someone's view about any randomly chosen political topic, and you could only ask them one question, what they think about race would be a good option.

The powerful influence of race on politics didn't end with Obama's presidency. Probably the most spirited debate around Donald Trump's political rise in 2015 and 2016 was whether his support was driven more by economic factors or racial resentment. During the campaign, Trump's speeches were full of racist, sexist, xenophobic, and otherwise incendiary comments. He demanded, for example, "a total and complete shutdown of Muslims entering the United States until our country's representatives can figure out what is going on."

Were voters supporting him because of or in spite of this talk? Many pundits and politicians argued for the latter, claiming voters were so concerned about pocketbook issues and the trajectory of the economy that they were willing to hold their noses and vote for Trump despite all the bigoted bluster. As US senator and presidential hopeful Bernie Sanders said, "Many of Trump's supporters are working-class people and they're angry, and they're angry because they're working longer hours for lower wages, they're angry because their jobs have left this country and gone to China or other low-wage countries, they're angry because they can't afford to send their kids to college so they can't retire with dignity."

But the truth is that this either/or framing—either economics or racism—presented a bogus choice. The economic grievances were *themselves* shaped by racial resentment. Trump supporters perceived themselves, and their peers, to be at greater economic risk than did non-Trump supporters, even when their economic positions were basically secure. In 2017, 86 percent of pro-Trump white millennials were employed, and they were less likely to be low income than anti-Trump white millennials. Whether or not a person described themselves as feeling economically anxious was better predicted by measures of their racial resentment than by any objective economic indicators.

Race is nearly everywhere in the United States. This is one reason why slogans like "policy change over mental change," which imply a choice between changing systems or changing hearts and minds, are misleading. When Americans try to figure out what to believe about some issue—whether to support healthcare reform or whom to vote for or what to think about climate science—their racial stereotypes and prejudices are right there, part of every decision. If prejudice drives economic anxiety (and economic anxiety reinforces prejudice), then it makes little sense to debate whether support for far-right politics is really driven by one or the other.

What this piece of either/or thinking misses is that for nearly any systemic problem in American political life, race sculpts public opinion about it. If there are six degrees of separation between each of us, there are at most two between race and everything else. The upshot of this, which we can call *two degrees of separation from race*, is twofold. On the one hand, critics of individualist approaches to racism are right to call for changes to the basic norms, laws, and structures maintaining inequalities between white and non-white people. On the other, the pursuit of those structural changes is no alternative to trying to change people's prejudices. Rather, those prejudices are themselves a central source of resistance to systemic change. The route to changing systems goes *through, not around, biased minds*.

Devine Intervention

But what if American hearts and minds can't, or simply won't, change? What if American racism is a habit too well inscribed in prevailing social structures to dislodge? As we explained in chapter 2, most antibias DEI workshops are thinly veiled attempts to insulate organizations from lawsuits and maintain the status quo. But not all of them. Some really do change minds.

Patricia Devine—who runs the Prejudice and Intergroup Relations Lab at the University of Wisconsin-Madison—isn't interested in helping people check “do something about racism” off their to-do list. “If we're just making white people feel better,” she says, “who cares?” Devine is the kind of person who commands an audience's attention. When she runs antibias workshops, she paces the room, making eye contact with each participant one by one. “I submit to you,” she is known to say, “that [prejudice is a habit that can be broken](#).” This kind of statement, said with conviction, can contribute to its own truth. Believing that you can break a habit is part of breaking a habit.

Devine has been refining her habit-breaking approach to prejudice for over thirty years. Her work flies under the radar. It's not flashy. Yet real, repeatedly replicated evidence suggests it works. It empowers people to make meaningful changes to their daily lives—and even, as we'll eventually see, their institutions.

Devine focuses on three things: belief, desire, and know-how. First is getting people to accept that there's a problem. They need to understand that they might be biased when they interpret someone on the street as looking dangerous or when they choose a partner for a class project.

Second, people need to be motivated to change. Devine aims to engage people who already want to be unprejudiced. She's not out to convince avowed racists of the wrongness of their ways. But even people who personally disavow racism may not be motivated to do much about it. They might think it's not *their* problem and that as long as they're cordial with everyone they meet, they're doing their part.

Devine's way of getting people to care is to convince them their biases aren't their fault—but that those biases cause harm nevertheless. Since stereotypes are peppered throughout American culture, everyone, including children, are repeatedly exposed to them. They're picked up early, like an accent. “[Biased responses become the default](#), automatic, and habitual reactions to members of stereotyped groups,” Devine writes, with her frequent collaborator William Cox. Putting it this way helps her tap into people's motivation because it defuses their defensiveness. The message is that people are not prejudiced because they have bad hearts. People are prejudiced because they live in a prejudiced world.

Conveying this message requires a subtle balancing act. Devine wants to get past her audience's defensiveness but also wants them to feel a little uncomfortable. The aim is to create a kind of tension—or “[cognitive dissonance](#)” as psychologists call it—as people recognize that their behavior is out of alignment with their values. Once they feel this kind of conflict—one inner voice saying that stereotyping is wrong, another saying that they, like everyone, are prone to stereotyping—people basically have three options. They can change their values, essentially saying “well if I act biased, it must be for a good reason. Maybe some of these so-called prejudices are justified.” Or they can reject the evidence: “maybe some people act in prejudiced ways but not me.” Or they can try to be less biased. Devine's goal is to guide participants toward this last option.

The third part of Devine's trainings imparts evidence-based lessons about how to change and how not to. For example, attempting to be “colorblind” doesn't work. Trying to “not see race” just makes race take up more attention. It's the analogue of an old experiment from social psychology, itself inspired by the novelist Fyodor Dostoevsky. If you're told “[don't think about a white bear](#),” guess what you just thought about? Instead, Devine teaches people to use strategies like “individuation,” where they focus on thinking about a person in terms of their individual characteristics, not stereotypes about their social group. Participants learn what to attend to rather than what to ignore.

This “[Devine intervention](#)” isn't going to change the world by itself. But carefully designed studies find that her participants become more aware of their biases, more concerned about discrimination, and more

empowered to take action, even years after the training. One study followed up with a group of Wisconsin undergrads two years after they completed Devine's program. Researchers posed as reporters for the school newspaper, *The Badger Herald*, and said they were testing a new editorial format: one student writes an opinion piece and another responds. The first opinion piece, "Racial stereotypes are useful tools," was already written. It argued that stereotypes were harmless and had only gotten a bad rap because of "PC culture."

The researchers asked the students, including one group who had completed Devine's program and a control group who hadn't, what they thought of the article. Were they interested in publishing a response? Pretty much everyone disagreed with the pro-stereotyping op-ed. [Yes, stereotypes are bad](#). No, we shouldn't promote them. But while just 48 percent of the control group opted to publish a response, a full 79 percent of Devine's former participants did.

What's the [secret sauce to the Devine intervention](#), when so many other approaches to prejudice reduction come up short? There's more than one ingredient. Her workshops give participants strategies they can adapt to their own situations, are based on well-established theories in cognitive and social science, and encourage people to become self-driven agents of change rather than simply telling them what to do. This is a far cry from Leaders' Choice Insurance (see chapter 2), where the cynical point of workshops is heading off lawsuits.

It's crucial to [distinguish good from bad workshops](#) and effective from ineffective leaders. Many antibias workshops *are* superficial, ineffective, and implemented for the wrong reasons. But acknowledging this doesn't lead to the conclusion that all forms of antibias education are a waste of time. Thinking otherwise would be like refusing to see any doctor because some doctors are on Big Pharma's payroll.

Mindsets That Matter

Critics say antibias workshops don't work. We agree they're right to worry, but the correct conclusion is that *poorly done* workshops don't work—and trainings implemented for cynical reasons are usually poorly done. When Trish Devine and like-minded scientists, teachers, and activists base their antibias efforts on solid science, keep at it for decades, and wear their genuine motivation on their sleeve, they can make a real difference. Given how racial bias colors people's perceptions of all aspects of modern life, this evidence for the malleability of the American mind is marvelous news. Getting more out of antibias education, however, requires seeing it from a both/and point of view.

An especially important lesson that good antibias workshops teach is that change is possible. In doing so, they foster in their participants a "growth mindset" toward prejudice reduction. Good workshops, in other words, emphasize the possibility of continuous incremental improvement, much like parents, counselors, and coaches often do. Whenever a coach says, "We're just trying to get a little bit better every day," they're invoking a growth mindset. The opposite is a "fixed mindset" that is centered on the idea that, for any kind of capability—musical talent, athletic potential, reasoning ability—people have what they have and not much is going to change.

Some worry that too much focus on people's "mindsets" is [another form of victim blaming](#) for larger structural problems. Others are skeptical that the effects of growth mindsets are as large as proponents claim. They suspect that structural factors affecting skill development like wealth, health, and racial privilege are much more important than individual mindsets. Their skepticism is buttressed by the finding that growth mindset interventions seem to work in some contexts but not in others. When researchers average across all the experiments, they get muddled results.

Fortunately, the next generation of studies is un-muddling the mess. One consistent finding is that teachers' mindsets matter too. The message of growth takes root when it's planted in the right soil: supportive teachers who genuinely believe their students can grow and who structure their classes in ways that encourage their students to prove them right.

When Devine looks workshop participants in the eye and insists that "prejudice is a habit that can be

broken,” she means it. Compare that to different experience: your company mandates a DEI training, but your boss says, “I know this training is a pain, but we just have to do it to get HR off our back.” So much for getting a little bit better today!

Studying and promoting growth mindsets is in no way inconsistent with focusing on the structures and environments in which people’s mindsets are embedded. The fact that [teachers’ mindsets matter](#) shows the opposite, that changing minds *requires* the right environment. The ways that teachers think, act, and structure their classes are vital parts of students’ environments—as anyone who has ever had an inspiring mentor knows.

Duck-Rabbiting Social Change

Promising as all this is, thinking of bias as a habit of mind might nevertheless seem trivializing. [Racism is a caste system](#), sustained by social structures like segregation. It’s baked into nearly every aspect of American life. In comparison to overthrowing white supremacy, an experiment that nudges a few more undergrads to clap back at a racist op-ed might look trifling. Is this a mere drop in a gigantic, structurally racist bucket?

We think this is a bad question. There’s no contradiction in thinking of racial bias as a habit and acknowledging its [rootedness in social structures](#). Nor is there any conflict between trying to change biases in individual minds and trying to abolish the structural conditions that spur them on.

Think about another bad habit: smoking. It can be tempting to frame smoking as a simple matter of individual vice, but it can also be understood as a structural problem. Cigarette companies lied for decades about the dangers of smoking. Dedicated sections in airplanes and [restaurants normalized it](#). Studies suggest that nicotine addiction is driven more by poverty than weakness of will; people living below the poverty line smoke nearly twice as long as those above it. Acknowledging that smoking is a habit doesn’t invalidate any of these points. And it doesn’t preclude paying attention to the structural factors that shape people’s habits.

Smoking is also a product of social networks. Kids start smoking when their peers do. As the economist Robert Frank has put it, the biggest danger of smoking in public isn’t secondhand smoke. [It’s creating more smokers](#) by making smoking seem normal and cool.

The good news is that the same social logic applies to quitting. The odds that people keep smoking drop by 25 percent when a sibling quits, by 36 percent when a friend quits, and by a whopping 67 percent when a spouse quits. Rates of smoking have plummeted from their peak in the 1960s and ’70s in large part because “whole groups of people were quitting in concert.”

Habit breaking, just like habit forming, is socially contagious. This is one of many phenomena that connect individuals to wider structures. It’s true that Trish Devine is trying to reduce prejudice one individual (or at least one workshop) at a time. It’s also true that one undergrad publishing one response to one racist op-ed isn’t the same as toppling the racial hierarchy. But Devine’s approach also leverages the fact that every workshop participant is a node in numerous social networks. Habits like smoking and prejudice spread through those networks and quitting them does too. Participants in the Devine intervention regularly report sharing what they’ve learned with others. And it looks like they can have a genuine impact on their peers. One study found that university professors who had not attended the training but worked closely with professors who had also become [more likely to work for social change](#) in their academic departments.

Thinking in a both/and way makes salient all sorts of external conditions that are important factors in change. The right teachers with the right mindsets, the right peers in the right networks, the right institutions with the right material and emotional support—these are all [essential ingredients of individual change](#).

But even if these trainings can, under the right conditions, reduce bias, what if there are other more impactful ways for people to direct their efforts to fight racial injustice? We turn now to what many critics

of individualist approaches to racism take to be a leading alternative to DEI workshops. As Bhaskar Sunkara, president of *The Nation* and founding editor of the socialist magazine *Jacobin*, says, “We do know, for example, of [a tool far more useful than unconscious bias trainings](#) in creating respect and equality: unions.”

Labor Organizing and Prejudice Reduction

Sunkara and others are onto something. Unions have a long history of promoting structural change. But these champions of unions are thinking about them too narrowly. Far from being competing alternatives, unions and antibias education are better understood as two mutually supporting pieces of broader struggles for justice.

Sunkara is right that unions do much more than advance their members’ financial interests. In some places, labor organizing has been integral to the struggle for racial justice. Martin Luther King Jr. saw the full threat of so-called “right-to-work” laws, for example, but put them in a broad context. These laws mandate that unions negotiate on behalf of nonunion members while [prohibiting unions from requiring](#) that those nonmembers pay dues. Right-to-work laws are part of the decades-long dismantling of labor power in the US. Anticipating their disastrous effects on economic and racial justice, King said, “In our glorious fight for civil rights, we must guard against being fooled by false slogans, such as ‘right to work.’ It is a law to rob us of our civil rights and job rights. Its purpose is to destroy labor unions and the freedom of collective bargaining by which unions have improved wages and working conditions of everyone. . . . Wherever these laws have been passed, wages are lower, job opportunities are fewer and there are no civil rights.”

King was prescient. In unions, people from varied walks of life can collaborate side by side as equals. This is one of the best ways to reduce prejudice. Social scientists have incorporated this insight into what they call the theory of intergroup contact. For people from different backgrounds—different politics, races, or religions—to form real bonds, they can’t just be thrown together in the same room. Anyone who’s seen a middle school cafeteria knows how easily we self-segregate. But when different kinds of people work alongside each other, doing something meaningful, on relatively equal status, their [prejudices tend to lessen](#).

It’s not just unions. Sports teams, first-year college roommates, and even Mormon missionaries have all been shown to reduce prejudice through intergroup contact. Unions are distinctive, though, because they’re not only places where workers have interracial interactions on relatively equal footing but also where collective bargaining creates incentives for union members to create strategic coalitions and cooperate across racial lines.

Consider again Sunkara’s claim about unions, antibias training, and prejudice. He’s right about the power of unions to reduce prejudice, but he’s wrong to promote unions rather than trying to change biases through workshops. What’s a good context for educating people about the importance of unions? One place is antibias workshops! Some—[like the ones Alex and Michael lead](#)—already do. Dedicating time in antibias workshops to the importance of unions is just one example of how these trainings can be oriented toward changing social structures. Workshops don’t have to only focus on our individual feelings and workplace interactions. They can also be *structure facing*. They can explain how individuals’ choices are shaped by social structures and guide participants toward ways to change them.

Structure-facing features are a centerpiece of Devine’s workshops. She begins by explaining how individuals’ prejudices result from growing up in a structurally prejudiced world, and then explores how to reconfigure those structures. For example, workshops with academic faculty consider different procedures for hiring and promotion. Participants learn about strategies like developing clear and consistent criteria for evaluating candidates ahead of time and sticking to them even when it’s difficult. Too often the alternative is that faculty fit the criteria to the candidates they like, who often happen to be a lot like the faculty themselves. As a result, successful candidates often end up being a lot like the white men who hire them. Devine’s workshops challenge these long-standing practices and encourage people to think through how to improve them.

Another of Devine's experiments, one that explores ways of overcoming gender discrimination, brings this point to life. Researchers worked with a range of science, engineering, math, and medical departments, again comparing departments that hadn't undergone her antibias training to those that had. Two years before the intervention, both sets of departments were hiring men over women at a rate of two to one. Two years afterward, the gender hiring gap was unchanged in the nonparticipating departments but had all but evaporated in the participating ones. What's more, faculty in the participating group (men and women alike) reported feeling that they "fit" better in their department, that their work was more valued, and that they were more comfortable talking about family obligations with colleagues.

Another way to put this is that people can learn to adopt [growth mindsets about institutions](#) themselves. Critics worry that promoting growth mindsets risks turning participants' gaze inward, to the thoughts in their heads rather than the systems that surround them. But there's nothing stopping workshops from also helping turn peoples' gaze outward. We can come to see the systems around us as malleable, not fixed in stone. The belief that the system can change contributes to its own truth.

Of course, just as antibias workshops are not inherently individualist, unions are not inherently prejudice reducing or structure changing. For example, some trade unions have histories of virulent racism and sexism. Police unions have blocked criminal justice reform and defended officers even in the worst cases of abuse. Mining, drilling, and fracking unions have been adamant [forces standing against clean energy](#). Sometimes unions reduce prejudice and help upend the status quo and sometimes they don't.

When they do, [what seems to make the difference](#) is how comfortable their members feel talking about race and politics, particularly around members of other racial groups. Acknowledging and confronting the issues helps; pretending they don't exist doesn't. The punditry seems increasingly convinced that antibias workshops are bullshit and that stubbornly racist unions won't improve without some kind of heavenly intrusion or deus ex machina. But maybe they just need a Devine intervention.

The Return of the Allegedly Irrelevant

If someone tells you to support unions rather than antibias workshops, suggest to them that better antibias workshops could strengthen unions. [Unions thrive when](#) members learn how race and racism are used to divide the working class. If someone tells you that the right way to fight racial inequality is universal healthcare instead of worrying about mere feelings and beliefs, remind them that prejudice stands in the way of universal healthcare. Years of racist backlash to Obamacare very nearly led to its undoing.

Similarly, if someone tells you that electing Democrats or turning back the climate crisis or increasing the minimum wage or ending capitalism is what it will take to conquer racism, remind them that reducing individual prejudice will be integral to each of these goals, too. Race and racism are as much causes of other social ills as they are effects.

We call this the Return of the Allegedly Irrelevant. Over and over again, we're told that efforts to make change, like attending an antibias workshop, are individualistic wastes of time, irrelevant to social change. We're told that the real, structure-changing action lies elsewhere, like in building unions and social movements. What we're not told, however, is that the allegedly irrelevant efforts often end up being vital for the supposedly relevant ones. The bridge-building needed for structure-changing unions and social movements goes through effective antibias workshops, or something a whole lot like them.

A Both/And Picture of People

Human Nature and Social Change

The impulse to frame problems and solutions in either/or terms is deeply rooted. It involves assumptions about human nature itself. To see how it's always both—changing systems is always a matter of changing people, and changing people is always a matter of changing systems—those assumptions need to be dragged into the light of day. Thankfully, a whole nexus of researchers from anthropology, biology, evolutionary science, and cognitive science has coalesced in recent decades around the idea that human beings are uniquely and spectacularly interdependent creatures. The social world permeates our inner worlds more than we realize. Far from being isolated individuals, detached from one another like solitary atoms, human beings are enmeshed in the wider world and with each other in surprising ways.

Similar ideas can be found in rich intellectual, spiritual, and political [traditions from around the world](#), including forms of Buddhism, Confucianism, feminism, environmentalism, and numerous indigenous ways of thinking, as we discuss below. Although there's not enough exchange, these schools of thought converge around a shared both/and picture of people.

WEIRD Culture

At least for Americans like us, one reason it's so easy to forget that it's always both is that we've been raised in what is probably the [most individualist culture of all time](#). We recounted the history of corporate PR that taught us to think of social problems like climate change and racism as matters of personal responsibility in chapters 1 and 2. A big part of why this PR worked is that American minds were primed for it. Think of macho John Wayne riding off into the sunset. Or Horatio Alger's fourteen-year-old hero pulling himself up by his bootstraps in *Ragged Dick*, aspiring "to turn over a new leaf, and try to grow up 'spectable.'"

The now familiar expression "[rugged individualism](#)" was coined by President Herbert Hoover in a 1928 campaign speech when he told voters they faced an either/or "choice between the American system of rugged individualism and a European philosophy of diametrically opposed doctrines of paternalism and state socialism." Hoover himself was appealing to an ethos powerfully articulated by Ralph Waldo Emerson nearly a century earlier in his essay "Self-Reliance." Emerson urged people to trust and stay true to themselves, to follow their own instincts and intuitions. He depicted social life as demanding conformity at the expense of authenticity, and counseled resistance: "Insist on yourself; never imitate."

Yet American individualism is just an extreme outgrowth of a much longer and more widespread set of cultural traditions. In a pathbreaking paper published in 2010, cultural evolutionists Joseph Henrich, Steven Heine, and Ara Norenzayan coined the acronym "WEIRD" to describe cultures that are [Western, Educated, Industrialized, Rich, and Democratic](#). The idea was that such cultures—especially in the United States—produce individuals with an idiosyncratic psychological profile. People in WEIRD cultures have come to associate rationality with thinking for oneself, for example. The motto of the Enlightenment, according to Immanuel Kant, was "Have the courage to use your own understanding."

WEIRD cultures also generate an image of people as egoists, driven by their own needs and desires. Governments are needed, in this picture, to keep human selfishness in check. "Government is necessary, not because man is naturally bad," Thomas Hobbes wrote in 1651, "but because man is by nature more individualistic than social." WEIRD cultures also tend to instill the belief that individuals know themselves better than they know others. Each person has privileged access to their own minds, and it's a typical WEIRD value to treasure authenticity—being your true self, whom only you truly know.

WEIRD minds are molded by WEIRD cultures, especially their norms, ideals, and institutions. [The “nuclear family” is a good example.](#) Picture a stereotypical suburban home, separated from the houses around it by a series of manicured lawns. Every kid in this home (all one or two of them, that is) has their own room, a private kingdom to personalize in whichever ways they choose to express their unique, authentic selves. Their house is tucked away in the urban outskirts, far from shared spaces like parks and libraries and Main Streets. Forget about bike lanes—the streets out here don’t even have sidewalks, so cars rule supreme. The parents drive off in separate vehicles to pursue their personal “callings,” which is a romantic spin on the fact that their job has become the all-consuming center of their lives. They spend whole days isolated in their office, often working through lunch. When they drive home again, they insist the household sits down together for fifteen to twenty solid minutes’ worth of the Family Meal. Before long, each WEIRDo retreats to separate rooms and separate screens, settling in to spend the rest of the evening curating their smartphone experiences, scrolling alone.

Of course, many American families don’t get to live this supposedly idyllic suburban life, but it’s incredible how many want to. Some version of this has been the American Dream since at least the 1950s.

As familiar as they might be, WEIRD minds, norms, and institutions are . . . weird. A wider look at familial, social, and cultural arrangements around the world, as well as back into our evolutionary past, shows that human beings are extraordinarily social animals. People are not at bottom isolated and egoistic; they’re gregarious and groupish. People are, and have evolved to be, thoroughly enmeshed with one another, saturated by the influence of other people, institutions, and culture. The old saw about “nature versus culture” is misleading. For human beings, our nature is cultural.

Human minds don’t start out as completely empty vessels, but a lot of what they end up containing is absorbed from their cultures and communities; it originates outside of them. In this way, we’re uniquely “outside-in” creatures. We soak up social structures exactly because we’re so thoroughly reliant on them, and our social worlds become woven right into the fabric of our minds and selves. The flip side of this is that while social influence is the air we breathe in, we’re always exhaling it too. Whether we realize it or not, we’re all [already equipped to be agents of social change.](#)

People Who Need People Are the Only People in the World

Henrich begins his book *The Secret of Our Success* with this thought experiment:

Suppose we took you and forty-nine of your coworkers and pitted you in a game of Survivor against a troop of fifty capuchin monkeys from Costa Rica. We would parachute both primate teams into the remote tropical forests of central Africa. After two years, we would return and count the survivors on each team. The team with the most survivors wins. Of course, neither team would be permitted to bring any equipment: no matches, water containers, knives, shoes, eyeglasses, antibiotics, pots, guns, or rope. To be kind, we would allow the humans—but not the monkeys—to wear clothes. Both teams would thus face surviving for years in a novel forest environment with only their wits, and their teammates, to rely on.

[Who would you bet on](#), the monkeys or you and your colleagues? Well, do you know how to make arrows, nets, and shelters? Do you know which plants or insects are toxic (many are) or how to detoxify them? Can you start a fire without matches or cook without a pot? Can you manufacture a fishhook? Do you know how to make natural adhesives? Which snakes are venomous? How will you protect yourself from predators at night? How will you get water? What is your knowledge of animal tracking?

The smart money’s on the monkeys, Henrich argues, “despite your team’s swollen crania and ample hubris.”

Even bringing along a boatload of equipment wouldn’t guarantee survival in an unknown environment.

In 1845, Sir John Franklin of England led the HMS *Erebus* and the HMS *Terror* in search of the Northwest Passage, a fabled waterway between Europe and Asia. Franklin's ships were state of the art, field-tested, reinforced icebreakers with steam engines and detachable rudders. They had coal-fired heaters, desalinators, and five years' worth of provisions. He expected to get locked in the ice in their first winter, which he and his crew did. They waited it out, as planned, and then moved south. Their second winter of being ice-locked was not planned, however. After nineteen months on the ice, the ships were abandoned in April 1848. Everyone on the voyage eventually died.

Henrich contrasts the fate of Franklin's crew to the indigenous Netsilik population who had been living in the area for tens of thousands of years. The comparison is apt. Netsilik communities were about the same size as Franklin's crew (around 100 people, though that includes children, whom Franklin didn't need to worry about).

What the Netsilik had and what Franklin's men lacked was the relevant cultural knowledge. The Netsilik built tents and kayaks out of driftwood in the summer and built snow houses on the ice during winter. They hunted reindeer and birds with sophisticated bows. They caught fish with three-pronged spears and seals with harpoons. In fact, there were so many big seals to hunt by the island's harbor that the Netsilik named it *Uqsuqtuuq*, or "lots of fat." In short, they found the same area—the one Franklin's crew found to be desolate and deadly—"rich in resources for food, clothing, shelter, and tool-making."

The Netsilik's cultural knowledge embodies what Henrich believes is the secret of human success in the animal world. The secret "resides not in the power of our individual minds, but in the *collective brains of our communities*." These collective "brains" are fundamentally social—made up of big groups of people knit together by a complex set of norms and practices—and fundamentally cultural too—containing centuries' worth of know-how passed along from one generation to the next.

Henrich and others argue that this feature of human nature, more than any other, enabled people to spread across the globe to inhabit virtually every land-based ecosystem—from deserts to rainforests to the arctic, from Polynesia to the Americas—long before they invented agriculture, cities, or writing. The human community has thrived despite its many comparative weaknesses to other animals: "*Any adult chimp can readily overpower us*, and any big cat can easily run us down. . . . Compared to other mammals of our size and diet, our colons are too short, stomachs too small, and teeth too petite. Our infants are born fat and dangerously premature, with skulls that have not yet fused." Likewise, human beings lack the kind of innate knowledge that allows other animals to instinctively meet survival needs. People need to learn, for example, which plants are poisonous and how to make fire and cook food.

What we're good at—when we act together rather than alone—is teaching, learning, and expanding our cultural store of knowledge, the information stored not in genes but in communities and practices. How this cultural knowledge grows and changes is at the center of an emerging scientific image of human beings. It's a profoundly different image from the individualist one many of us grew up with.

The Cultural Evolution of Human Nature

Aristotle was so impressed by our species' sociality that he claimed in his *Politics* that we are "by nature a political animal." "No quality of human nature is more remarkable," *the Scottish philosopher David Hume* wrote, "both in itself and in its consequences, than that propensity we have to sympathize with others, and to receive by communication their inclinations and sentiments, however different from, or even contrary to our own." Charles Darwin deserves a spot at the top of any list of towering precursors to the modern scientific image of our species too.

Similar themes are found in *the work of feminists* like Audre Lorde, Barbara Smith, and other authors of 1977's Combahee River Collective statement; Gloria Anzaldúa, Cherríe Moraga, and other self-described "third-world women" in the 1981 *This Bridge Called My Back: Writings by Radical Women of Color*; and many more. These scholars have long rejected the isolated, atomic view of the self, defined in terms of the private inner workings of the rational autonomous mind. In its place, they've described a dynamic and relational self, one constituted by its relations with other people.

Our relationships to others are literally part of what makes us who we are. Gertrude Stein put it beautifully: in lieu of Descartes' claim that "I think therefore I am," she wrote, "[I am because my little dog knows me.](#)" The Zulu and Xhosa notion of *ubuntu* captures a similar idea; often translated as "humanity toward others," it suggests that "I am because we are." Nelson Mandela praised ubuntu philosophy for capturing "the profound sense that we are human only through the humanity of others."

These ideas also anticipate much of what's emerging from contemporary work on human psychology and cultural evolution. Human beings are distinctively and completely reliant on each other: natural born social learners, eager collaborators, and cooperative norm followers. Emotionally, people crave togetherness and belonging. Cognitively, [human beings are equipped with sophisticated capacities](#) that allow them to navigate across many overlapping social worlds and find meaning in attachments and collective activities. People easily and instinctively thrive in dynamic networks of social relationships because human minds are built for it.

Like other species, *Homo sapiens* is a product of evolution. But in addition to the process of natural selection acting on genes, people are also shaped by various forms of cultural selection. The term "culture" has a technical meaning in this context. It picks out a class of information, namely information that is transmitted socially. The concept provides a useful contrast with the genetic information that is transmitted via genes. A bit of information counts as cultural when it is passed from one organism to the next not by genetic means but by social processes. All living things are endowed with a genetic inheritance, but human beings enjoy an extensive cultural inheritance as well.

Cultural information can be stored in brains but also outside of them: in artifacts, technologies, stories, books, traditions, institutions, built environments, and social structures. Culturally acquired traits like the desire for the American Dream aren't innate. They develop over time, piece by piece, with improvements and additions made by many people dispersed across space and time.

Socially transmitted information accumulates, allowing later generations to retain and build upon the knowledge, skills, and traditions of previous generations. Think about architecture, animal husbandry, or cooking—all are the product of slowly evolving and accumulating cultural knowledge.

Less useful ideas and behaviors are forgotten, discarded, or otherwise fall out of circulation, and as [more useful ideas and behaviors are adopted](#) by more individuals, they spread and develop over time. This is what the Netsilik had that Franklin's crew did not: a cultural inheritance that, over the course of generations, had slowly accumulated knowledge and skills and technologies that were specifically adapted to the challenges of surviving in the tundra.

Another idea central to this picture of human nature is that the two inheritance systems—the genetic and the cultural—came to interact with each other in an increasingly complex system of feedback loops known as "gene-culture coevolution." A classic example is lactase persistence in adults, which is what allows some adults to be able to digest milk. Milk contains a sugar called lactose, and digesting it requires the presence of an enzyme called lactase in the intestine. Almost all babies have lactase, while many adults don't. So what explains lactase persistence in other adults?

[Evolutionary geneticists have traced this trait](#) to ancient cultures that used milk for sustenance. It developed separately in distinct regions, including sub-Saharan Africa, Europe, and the Middle East, between 8000 and 4000 BCE. Descendants of people from these areas are now genetically disposed to be able to drink milk as adults, while those descended from areas that didn't use milk, including much of East Asia, the Americas, and Southern Africa, are not. In short, the cultural evolution of dairy farming drove the genetic inheritance of lactase persistence. Culture changes genes. But gene-culture coevolution is a feedback system. The genetic ability to tolerate milk as an adult in turn shapes cultural practices, which bring you Parmigiano-Reggiano and the shmear of cream cheese on a bagel. Just as people shape systems and systems shape people, genes shape cultures and cultures shape genes.

Gregariousness

Earlier we noted how, if dropped off in a tropical forest, a group of fifty capuchin monkeys would be more

likely to survive than a group including you and forty-nine of your best friends. But in other circumstances, the odds flip. Consider the primatologist Sarah Hrdy's thought experiment, which she called "[apes on a plane](#)":

Each year 1.6 billion passengers fly to destinations around the world. Patiently we line up to be checked and patted down by someone we've never seen before. We file on board an aluminum cylinder and cram our bodies into narrow seats, elbow to elbow, accommodating one another for as long as the flight takes. With nods and resigned smiles, passengers make eye contact and then yield to latecomers pushing past. When a young man wearing a backpack hits me with it as he reaches up to cram his excess paraphernalia into an overhead compartment, instead of grimacing or bearing my teeth, I smile weakly, disguising my irritation. Most people on board ignore the crying baby or pretend to. . . .

I cannot help but wonder what would happen if my fellow human passengers suddenly morphed into another species of ape. What if I were traveling with a planeload of chimpanzees? Any one of us would be lucky to disembark with all 10 fingers and toes still attached, with the babies still breathing and unmaimed. Bloody earlobes and other appendages would litter the aisles. Compressing so many highly impulsive strangers into a tight space would be a recipe for mayhem.

Individualist cultures depict human beings as fundamentally self-interested, predisposed to be on guard against one another. But in reality, while people are sometimes selfish and suspicious of each other, these disruptions of trust occur against a backdrop of otherwise massively well-organized, unthinkingly efficient, and stunningly smooth cooperation.

Human beings in general are not just social but also abnormally socially tolerant. We're temperamentally more docile and less aggressive than even our most closely related species. There's a lot of variability, of course. Some people are bookworms, while others are the life of the party. Most are peaceful, but some are aggressive and violent. On the whole, though, human beings have a unique and [unparalleled capacity for cooperation](#). People tolerate strangers, trade with rivals, and usually go about their business without confrontation, even when others are annoying, insulting, or threatening.

This thinning out of aggression from human nature, sometimes called "self-domestication," is tied to our species' uniquely deep entanglement with culture. In so many ways, people's access to what they need—from resources to recognition—depends on playing nicely with others. They've got to read the room. This gregariousness permits individuals to get and stay close to each other, to knit together into families, bands, and larger and more complex communities. These social arrangements in turn require more and more ability to collaborate, keep track of social networks, and understand each other's minds. In many ways, the story of human evolution is the story of culture and cooperative psychology ratcheting each other up in a feedback loop.

People *are* sometimes selfish or vicious, but these are not humanity's fixed, core traits. Human beings are outfitted with altruistic emotions like empathy, sympathy, love, and compassion. We're also equipped with more evolutionarily recent, [distinctively human social emotions](#): shame, guilt, embarrassment, social anxiety, and moral disgust. These emotions help keep people obsessed with rules and norms, closely monitoring who's following them and who's breaking them. Gratitude, respect, team spirit, group pride, and patriotism are all affiliative too, even if they sometimes have dark sides. Even experiences like confusion, boredom, and curiosity help guide people through the enormous flow of cultural information they encounter every day.

The individualist view that self-interest is the fundamental motivational core of human nature isn't just misleading but also dreadfully flattening. Mountains of evidence tell us that it's time to replace this oversimplification with a picture of human nature that is richer, more complex, and more outside-in,

highlighting the relationships that connect and shape us. So think twice the next time someone suggests that “change begins within.” The changes within us are inseparable from the people and structures around us.

Interdependence

Imagine showing a four-year-old kid a puzzle box with a sticker inside. Opening the box requires a series of steps. There’s a tool that allows you to move a bolt, which allows you to slide the tool into a hole, which allows you to open a door, and so on. You tell the kid that they can try to open the box however they want. Then you demonstrate how to open it and then leave the room. A hidden video camera records what the kid does.

It’s no surprise that most kids imitate the steps they were shown to open the box. But there’s a twist. When demonstrating, you’ve slipped in a couple unnecessary steps, like tapping the tool against the side of the box. Do the kids imitate these unnecessary steps too? You bet. They’re doing what’s called “over-imitating.” It might sound cute (and it is), but it’s also a window into something distinctive and deeply important about us.

In a famous study of over-imitation—the one we’ve just described—most four-year-old kids copied all the researcher’s steps no matter whether the box was transparent or opaque. When the box was see-through and they could tell that tapping its side did nothing to help them open it, the kids still copied the unnecessary steps. This is the “over” part of over-imitation, and human children appear wired up for it. Over-imitation has been found all over the world, even in the Bushmen of Africa’s Kalahari Desert, a culture with little tradition of explicit teaching. There, kids follow their elders around, soak up what they see, and build skills through trial and error. Still, researchers have found that they over-imitate the adults around them.

Even more striking, in this study and others, is the comparison between how kids and nonhuman animals perform. In an experiment comparing four-year-olds to chimpanzees, both the kids and the chimps were shown necessary and unnecessary steps for opening the box. In contrast to the kids, the chimps skipped the unnecessary steps when they could see that they didn’t do anything. “Animals focus on getting the job done,” as Mark Nielsen, a psychologist who studies comparative cognition put it, while “humans seem to almost forget about the outcome and copy everything we see.”

Research like this—comparing imitation in human and nonhuman animals—can make human beings look silly. Chimps, dogs, and elephants have been shown to solve certain tasks more efficiently than human children, who will go through all kinds of elaborate needless steps if that’s what they’ve seen other people do. But while examples of over-imitation can seem mindless in isolation, this tendency for conformity has a good rationale.

Viewed from the perspective of the cultural evolution of human nature, over-imitation is a feature, not a bug. It functions to help ensure that people make efficient use of the vast wisdom found in culture, all those complex skills and technologies and institutional procedures pieced together and refined by generations of forebears. The instinct to copy is so strong because learning too much from others is a “better” mistake than copying and learning too little—especially for a species that relies so heavily on culture. Following the wisdom of the collective is usually a better bet than relying on the intelligence of the individual.

Reasoning like this can recast other aspects of human psychology. A “conformity bias” makes it feel sensible to do what seems normal, and a “prestige bias” inclines people to copy those who have found success. People tend to be relentlessly interested in what others are thinking and have a unique set of tools for “mentalizing”—predicting what others will do and trying to understand them. Instincts like these make it easier to learn from others, to learn the right things, and to execute what’s been learned correctly. They allow better and smoother access to the cultural inheritance system.

All these instincts lead groups of people to accumulate cultural knowledge over time, even when the individuals who use it don’t know exactly what it’s doing. Leading up to the Viking Age, sturdy iron was

hard to find in the Scandinavian bogs. So warriors sought out another way to compensate for the weakness of their iron swords: infusing weaponry with the spirits of the dead. They mixed in the bones of their ancestors and animals. What they were also doing—unknowingly—was turning iron into steel. Much like making charcoal out of wood, burning bones in oxygen-poor conditions creates “bone coal.” Their swords, axes, and arrowheads became much stronger but not for the reasons they thought. It wasn’t individual rational problem-solving that led to the creation and persistence of this ritual; it was cultural evolutionary processes that involved over-imitation, conformity bias, prestige bias, and social interdependence.

Interdependence for Social Change

Compared to the individualist image of human nature that holds so much sway in the WEIRD world, this alternative culture-centered image is good news for anyone interested in social change. It softens the perceived boundaries that separate people from each other as well as those between people and the systems they live within. Human beings are designed by evolution to inhabit their world but also to shape and reshape the structures that organize it.

This image of individuals and structures as deeply interwoven is a key piece of a difficult puzzle, the puzzle of figuring out how to see ourselves as part of “big structural change.” People are more influenced by their surroundings than we notice, but there’s also a flip side to this insight. We’re not just the over-imitators; we are the over-imitated.

We Are Each Other's Situation

Recycling Revisited

"I work in the environmental movement. [I don't care if you recycle](#)," wrote the climate justice activist and journalist Mary Annaïse Heglar in 2019, "Stop obsessing over your environmental 'sins.' Fight the oil and gas industry instead."

Individual actions like recycling can indeed seem trivial when they're stacked up next to taking down the fossil fuel industry. But from a both/and perspective, recycling turns out to be an excellent case study of how we can use our radical interdependence for good.

People are used to thinking that what matters about their actions are the immediate material consequences, like how many bottles and cans they throw in the blue bin and how big their overall carbon footprint is. But climate scientist Katherine Hayhoe points to another kind of consequence: "[our personal actions do matter](#), but they matter because they can change others." This chapter unpacks Hayhoe's idea. Our actions affect other people in lots of ways, including by shaping social norms. What's more, one of the "other" people your personal actions can affect is yourself—that is, yourself in the future. Little things you do now can put you on the path to doing more and bigger things later.

It's odd that so many people's first inclination when they think about fighting climate change is to get serious about recycling. Landfills do contribute to global warming, but they're one cause among many. Recycling owes much of its outsized prominence to the corporate PR campaigns about individual responsibility we saw in chapter 1. Recycling was also emphasized by the environmental movement in its focus on wastefulness.

When it comes to plastic waste, there really are reasons to worry. Plastic waste is contributing to species extinction, ecological devastation, and numerous health problems. [Microplastics are found](#) in the Pyrenees Mountains over 100 miles from the nearest town, in fields where food is grown, in virtually all brands of bottled water, and in our bodies. Plastics contribute to climate change more directly, too. Ninety-eight percent of them are made from fossil fuels, and as energy production shifts to renewable sources, corporations are using more and more fossil fuels to make plastic. If every other source of carbon pollution in the world was cut to zero tomorrow, plastic production alone would heat the Earth 1.5°C as early as 2060.

Given all this pollution, it's reasonable to try to recycle. In the US, about [68 percent of paper gets recycled](#), and Americans recycle a lot of metal and glass too. Unfortunately, plastics recycling is a disaster. As one set of experts put it, when it comes to plastic, "recycling does not work and will never work." There are thousands of types of plastic, each requiring a distinctive recycling process. And unlike glass and paper, plastic isn't inert. It absorbs all kinds of dangerous chemicals, which limits the uses of recycled plastic. Even the process of recycling plastic is wasteful and expensive.

As a result, only about 8 percent of American plastic overall gets recycled. Americans used to ship vast amounts of it to China, where it was "recycled"—which usually meant it just got burned or buried. Then in 2018, China prohibited most plastics importing. In response, the US started shipping tens of thousands of containers of plastic to Malaysia, Thailand, and Vietnam instead. Those countries quickly banned plastic waste imports too, in part because burning it contaminated water, killed crops, and inflamed respiratory illnesses. So the US turned to Bangladesh, Cambodia, Ethiopia, Ghana, Kenya, Laos, and Senegal.

The US does its fair share of plastic incineration as well. Americans burn six times more plastic than they recycle. Where, you ask? Once again, everything is two degrees separated from race. The United States' most marginalized communities, the ones with more poverty and fewer white people, [house the](#)

overwhelming majority of plastic incinerators (nearly eight out of ten, according to one 2019 report). The proliferation of these incinerators is a catastrophe as well as an injustice—in this case, what has become known as environmental injustice.

For all these reasons, when a company like Starbucks pats itself on the back for transitioning to “recyclable” lids, they’re doing little more than buying themselves a bit of good publicity. Only 5 percent of polypropylene—or the number 5 plastic making up Starbucks’s lids—was recycled in 2015. (Alex’s sunny Southern Californian city of Claremont doesn’t even pretend to recycle any plastic higher than number 2.) Nearly all the lids end up in the same places as most other plastic: [in the ocean, the sky, our food, your lungs](#).

With all this in mind, should you bother recycling? In public discussion of this question, the answer has a standard—and by now familiar—format. It goes something like this:

Expert: Individuals can’t fix this problem. We need to change the system.

Journalist’s silent plea: But we have to give the reader something to do.

Expert: Here are some small steps you can take [to feel like you’re doing your part](#).

There *are* structural fixes for this problem. One approach—familiar from the early days of the American Can Company and its fight against disposable bottle bans—uses “producer responsibility” laws. Like a carbon tax, these laws shift the cost of recycling to manufacturers by taxing products with high environmental impacts and offering incentives to those who produce “greener” items. By 2022, [Maine, Oregon, and California](#) had all passed laws like these. When there’s a market for recycled goods, manufacturers themselves can end up supporting such laws.

How do we get from here, where plastic is a disaster, to there, a place where the markets for recycled goods are large, single-use plastics are no more, and producer responsibility laws are common? Another duck-rabbit shift in perspective will be valuable in finding a way. Individual actions like recycling can be measured not just by their material consequences—how many bottles you save by doggedly sorting the trash—but also by their social consequences.

The Power of Moral Charisma

Understanding social influence requires flipping the script on a long-standing psychological debate. For decades starting in the 1960s, psychologists studying personality and behavior were locked in an argument. One side took what seems like the obvious position: people have personalities, and those personalities matter. Traits like being funny or shy or optimistic are real and contribute to people’s character. “Personality psychologists” measured traits like these and used those measurements to make predictions about people’s behavior. If a person is shy, they’ll be less likely to start a conversation with a stranger. If someone is funny, they’ll tend to tell jokes and make people laugh.

This seemingly uncontroversial line of thought ran into problems. When personality psychologists measured traits of character and then tried to predict actual behavior—starting conversations, telling jokes—the measurements didn’t do a stellar job. Someone who scored high on a test of honesty wasn’t much more likely to avoid telling a convenient lie than someone who scored low on a test of honesty. Ostensibly stingy people donated just as much money to a stranger in need as the ostensibly generous.

A rival view, which came to be known as “situationism,” emerged to explain these surprising results. Proponents of situationism argued that personality is basically a myth. People aren’t carrying around within themselves stable traits like generosity and honesty. Instead, the situation we’re in largely determines what we do. In some situations—say, when the penalties for getting caught lying are steep—people tend to be honest regardless of who they are. In other situations where the stakes are low, people of all putative personality types tend to fudge the truth.

The situationist view has its own intuitive appeal. It too has a claim to being rooted in common sense,

where it's expressed in ideas like "peer pressure" and platitudes like "power corrupts." Put most people near enough to power and they'll lie, cheat, and steal. On a more day-to-day level, people can be honest with their boss but not their kids, or be kind right after getting a compliment but grouchy after getting insulted. The situationist charge was that people's behavior is so fluid, so context-bound, that it's wrong to think of behavior as driven by internal character traits at all. People are much more like grass blowing in the wind than personality psychology suggested.

The academic debate cooled off after a few decades when it became clear that both sides were kind of right. There's little doubt that many popular ways of measuring personality are at best entertainment and [at worst pernicious scams](#). The Myers-Briggs Type Indicator (MBTI), which divides personality into sixteen types, is exhibit A in the courtroom of bullshit. It's as scientific as zodiac signs or a Hogwarts House Test. But improvements in the actual science of personality, along with new statistical techniques for measuring personality over time, have produced better results.

Scientifically validated traits include the Big Five, which go by the acronym OCEAN: Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. Openness, for example, stands for "openness to new experience," and people who rate high in it are more likely to try exotic foods, travel on adventures, and feel comfortable talking to randos. Conscientious people tend to be self-disciplined, preferring plans to chance encounters and rule following to rule breaking.

There is a legitimate (albeit imperfect and ever-evolving) science to personality. But even using these newer, well-validated measures, nobody is open to new experience or conscientious or anything else all the time and independently of context. Even those very open to new experiences might not like spicy food, so when their friends suggest trying the new Thai restaurant down the street, they decline. They can have the stable trait of being closed to new experiences with spicy food but have another stable trait alongside it: openness to nonspicy foods.

The name for this view—that [personality is both real but also highly situation dependent](#)—is "interactionism." Traits of character are the product of interactions between features of persons—things that people are born with or take shape in early childhood—and features of the environment—that situational "wind" blowing us around.

Even with this advance, something was still missing from these discussions about personality. Hagop Sarkissian, a philosopher who studies both social psychology and classical Chinese ethics, noticed in the early aughts that acknowledging the power of situations seemed to leave little room for individual freedom. People talked about "the situation" like it was some kind of faceless force outside individuals, controlling their choices. But Sarkissian stressed that one of the most important elements of the situations shaping people's choices is other people. We are each other's situations.

Recall, for example, that the best way [to predict whether someone will become a smoker](#) is by looking at how much time they spend around other smokers. The power of situations, as Sarkissian put it, is reciprocal. It creates a looping effect. If you're part of my situation, shaping my choices, then I'm part of yours too. The fact that smokers help make other smokers shows how we're all social influencers.

As Sarkissian put it, "we do not simply *react* to external situations, but [we also shape our situations](#) through the variables we ourselves introduce." His work develops this insight by connecting it to the virtue of *de* () in classical Confucian ethics, which is a kind of moral charisma. Individuals who cultivate it influence others by shaping their shared situations.

Confucian ethics sweats the small stuff. It directs people to pay meticulous attention to their mannerisms, tone of voice, posture, and expressions. [The Analects of Confucius is packed](#) full of instructions about how to bow to this person or that, what color clothes to wear for one occasion or another, and so on. For example, Confucius's ideal leader "wears a black robe over a lambskin coat; a white robe over a fawn fur coat; and a yellow robe over a fox fur coat."

Confucius compiled a cookbook for using human interconnectedness for good. Many of his recipes, like the one prescribing which fur to wear for which occasion, are specific to his place and time, but they're easy enough to translate. Minding our manners is another way we influence each other, making etiquette [a](#)

vital part of moral charisma. Sarkissian's spin on the Confucian virtue of *de* points to the potential for change rooted in humanity's outside-in nature (chapter 6). Each person is the powerful outside to someone else's inside. The same is true when it comes to helping each other change systems and structures.

Social Norms for Social Influence

We all help make up each other's situations. This means that even seemingly individual choices aren't so individual. There are also better and worse ways to be someone else's situation. A recent explosion of research on social norms is helping to distinguish the better from the worse.

Social norms are **unstated rules about "what we do."** They're everywhere, telling Americans to shake hands but Japanese people to bow. The norms of other places and times can seem strange when viewed from afar, like the Victorian norm to never ask a direct question. (A polite person would say, "I hope you're doing well" but never ask "how are you doing?") Norms are also often imbued with prejudice and stereotypes. Women in the 1950s were instructed to avoid letting their hands hang at their sides "like dead fish," as it would detract from the silhouette of their figure and make them look "lifeless" (figure 7.1).



"Dead Fish"

Figure 7.1

"Dead fish" hands.

Norms aren't just ubiquitous; **they're powerful.** Far from merely informing people about, say, which fork is for the salad, they're essential for the kind of cooperation we explored in the last chapter that makes the human species so distinctive. Norms help people solve collective action problems, situations where it's in everyone's collective interest to cooperate but no one's interest to do so alone. They're indispensable to the effectiveness of governments, affecting how policies and laws are actually implemented. Which norms prevail in a polity can make the difference between tolerating corruption and uprooting it.

Knowing the norms of a community is part of what it is to be a member of one. In the evolutionary past, in order to build things together, trade, and intermarry, people needed to know what to expect from one another without relying on governments to lay down the law (since governments didn't yet exist). Norms provided this information.

But norms don't *just* provide information. They don't merely encode what "people around here do." They come with a complex motivational oomph. In grasping norms—internalizing a community's unwritten rules—we become motivated to follow them. We also become ready to enforce them and to

accept it when others enforce them on us. Have you been stuck in line at the supermarket behind someone talking loudly on speakerphone? There's no law against this, but that person is apt to get a lot of side-eye. That's [norm enforcement in action](#).

Research on children shows that the motivation to learn, follow, and enforce norms emerges at a remarkably young age. As soon as most children learn “a blickett is for twisting” (or whatever norm for using a new toy the researcher teaches them), they're ready to lay down the law against would-be renegades [who try kicking their blicketts](#). They'll start to internalize the norm straightaway, and once they do, they won't need encouragement to follow it anymore. Nor will they need encouragement to convey their disapproval to anyone else who fails to comply.

Even when norms are arbitrary—like “twist blicketts,” “use the small fork for salads,” and “don't ask direct questions”—following and enforcing them becomes, psychologically, something like an end in itself. Those people in the supermarket line are monitoring your behavior, ready to react negatively if, say, you're chatting away on speaker instead of putting your food on the conveyor belt. You may not always realize it, but you're doing the same to them. You might not love it when others “police” your checkout etiquette, but you've almost certainly policed someone else's. This can seem trivial and annoying, but social norms are what allow us to [Live In A Society](#).

Putting all this in the context of creating social change, Michael Grunwald returns to the iconic Pogo poster, which announced that “we have met the enemy [and he is us](#).” The poster served the interests of industry, but it also “helped make littering extremely uncool.” Grunwald continues:

Carbon emissions are atmospheric litter. Every gram of carbon makes global warming a bit worse. But carbon litter is invisible, while normal litter is ugly, which helps explain why carbon litter, though much harder to remove than normal litter, is not yet uncool.

As solar goes mainstream, fossil electricity could become as uncool as littering; as electric cars get cheaper, the same thing could happen to internal combustion engines. The Crying Indian illustrated how an individual ethic can be as contagious as a virus, inspiring people to do things because they just seem like the thing to do.

We think Grunwald is spot-on. Researchers from across academic disciplines have explored ways to take his ideas further, to use our position as each other's situations [to scale up the potency of personal choices](#).

Putting Norms to Work

It can be pretty tempting to join in when “everyone's doing it,” whatever “it” is. This pull drives social change. According to some estimates, nothing moves us to get involved more than [comparing our choices to others' choices](#). Likewise, people who get involved in collective movements—as we'll dig into in the coming chapters—tend to feel highly motivated to follow the movement's norms.

This is especially true when people get a sense that things are changing, that what “we normally do around here” is in flux. Researchers think about trends on the brink of catching on in terms of “dynamic” norms. Early adopters of dynamic norms can enjoy the pleasure of getting to say “[I was onto that before it was cool](#)” rather than being the last to hop on the bandwagon. Take a headline from *The Hill*: “Vegetarianism Is on the Rise—Especially the Part-Time Kind.” Telling readers how “more and more” people are reducing meat consumption can be a more powerful driver of norm change than describing the stats as static (i.e., simply stating how many people avoid meat at a snapshot in time).

This kind of messaging is particularly helpful for promoting new norms when they haven't yet managed to go mainstream. Political scientists have documented [many cases of tipping points and “norm cascades,”](#) when early uptake of a new norm is slow but then a critical mass of support is reached, triggering a sudden and rapid spread. (An unparalleled pandemic example from the beginning of the COVID-19 years: “masks

on!” Then, “masks off!”) Again, much of the power of norm-based approaches to social change is that learning norms involves more than gaining information about what people are up to. Even just learning about an emerging norm can galvanize action.

There are also benefits to talking about the importance of a norm out loud, to consciously and explicitly announcing one’s embrace of the norm, both to oneself and to others. It’s one thing to silently intend to go to the gym and another to stake one’s reputation on being a gym person by talking with friends about how good it feels to exercise. In conspicuously supporting a norm in public, people help spread it through the community. They become norm entrepreneurs.

Central to the power of norms is how they are enforced by the community, but norm enforcement doesn’t only take the form of punishing violations. Praise is critical to norm change too. People help promote new norms—like bicycle commuting and getting vaxxed—when they [applaud others for adopting them](#). In return, the people who express praise feel good about supporting other members in the growing club. Praise offers the praiser a little do-gooder booster shot.

Social Influence on Your Future Self: How Each Choice Affects the Next

To summarize: the power of individual action goes beyond tallying up material consequences, like how much less gas gets guzzled when one driver switches to an EV. It also includes social consequences, like the way a single norm entrepreneur can shape others’ situations. But the power of individual action goes further still. There are consequences for our future selves. The choices we make today affect the person we become tomorrow.

There’s a familiar and depressing version of this: people sometimes do good deeds to get away with bad ones. Some of the most famous examples are the most atrocious. In the late 1800s, while oppressing, mutilating, and massacring African colonial subjects in what today is the Democratic Republic of the Congo, the [Belgian King Leopold II won acclaim](#) for his philanthropic work. The International Stalin Prize for Strengthening Peace Among Peoples (aka “The Lenin”) was created not long after Stalin starved 3–5 million Ukrainians with his agricultural collectivization plan. In 1988, Bill Cosby donated \$20 million dollars to Spelman College, the most prestigious historically Black college for women in the country, at the same time as he was well on his way to allegedly raping sixty women. “The great mystery of evil,” Jelani Cobb wrote, “is not that it persists but, rather, that so many of its practitioners wish to do so while being thought of as saints.”

Non-evil people are susceptible to a more banal form of the same dynamic, making this susceptibility a cause for alarm about individualist approaches to social change. What if doing a good deed now—say, buying solar panels—makes a person feel “licensed” to do less for the climate going forward? What if people think they can “make up” for their political inaction through superficial gestures toward personal improvement?

It’s an understandable problem. Take someone who cares deeply about the environment. They believe the most impactful personal changes include flying less, driving less, and eating less meat. Now suppose they’ve just taken a twelve-hour overnight train ride to visit family, foregoing what would have been a two-hour flight for a comparable price. They walk in the door to the smell of sizzling bacon in grandma’s kitchen. They’ve been keeping things pretty plant-based at home, but, hey, what’s the harm in eating a few strips? After all, they just did their part to curb carbon pollution by taking the train. Doing all that good gives them license [to get away with a peccadillo or two](#), especially if they wait to eat that bacon when no one else is looking.

What’s happening here is that one choice has downstream influence on another, with the first choice spilling over to affect the next. After making one climate-friendly choice, a person might feel less guilty about a subsequent climate-unfriendly choice. This is called “negative spillover,” when doing something good makes people more likely to turn around and do something bad.

Critics of individualist approaches to climate action often invoke the threat of negative spillover (even if they don’t use the specific term). There are two distinct but related concerns. One is that people’s efforts to

make green choices are sometimes canceled out because they feel permitted to engage in even more carbon-intensive behaviors. For example, studies have found that [people who weatherize their homes](#) tend to raise their thermostats, ironically resulting in decreased energy savings and increased pollution.

The second concern is that expending energy on things like flying less or weatherizing one's home is a diversion. Critics say that individual sacrifices like these are a misuse of time, energy, and resources better spent on more important opportunities to effect larger-scale structural change. Mary Annaïse Heglar writes, "The belief that this enormous, existential problem could have been fixed if all of us had just tweaked our consumptive habits is not only preposterous; it's dangerous. It [turns environmentalism into an individual choice](#) defined as sin or virtue, convicting those who don't or can't uphold these ethics."

The idea is that tweaking our individual consumption habits takes away from more important—political, impactful—things we could be doing. It can also lead us to blame people for problems that aren't really their fault. As we detailed in chapter 1, the whole impetus behind corporate campaigns against littering and for carbon footprint monitoring has been to keep the public complacent and distracted from big business' much larger contributions to ecological devastation and the climate crisis.

But does it follow that people should focus their efforts on making these big projects happen *rather than* worrying about personal consumption? Is the personal stuff a counterproductive distraction from joining the larger battle?

No, or at least not always. Spillover can also be positive. Start with some simple, smaller-scale examples at the personal level. Hotel guests who make a commitment to reuse their towels are more likely to turn off the lights when they leave the room. Californians who start composting go on to [save more energy and water](#).

In these examples of positive spillover, the values expressed in one decision carry forward and positively reinforce decisions about other things in the future. One good deed for the climate can beget the next. In at least some cases, the effect here looks genuinely causal: the first choice drives the second. Reusing towels causes people to turn off the lights, and composting makes people more likely to save energy and water.

One might worry, though, that these examples of positive spillover all happen between similar kinds of choices. They all have to do with personal consumption choices. What about positive spillover from the personal to the political?

In research on voters in Maine, John Thøgerson and Caroline Noblet found that personal behavior changes, including buying energy-efficient products and, yes, even recycling, made people [more supportive of wind power projects](#). Their findings suggest that small personal steps can springboard people into backing substantive structural change. Taking action may have helped shape people's policy preferences over and above how committed they already felt to environmental issues. So it wasn't just that people who cared about the environment were more likely both to be "green consumers" and to support wind power. It looks like people who acted like green consumers *became* more concerned about the environment—and also more supportive of wind power.

Studies like this give reason to think that personal behavior changes don't necessarily crowd out actions focused on structural change. They can crowd them in too. Thøgerson and Noblet think of positive spillover as a kind of catalytic conversion, where seemingly small, inconsequential things have the potential to catalyze interest in doing other, more potentially impactful things.

Research on positive spillover is young but suggestive. One working paper for the National Bureau of Economic Research finds that German households that put solar panels on their roofs become [more likely to vote for the Green Party](#). It's not just that people who put up panels are already likely to vote Green. There seems to be a causal effect: installing the panels subsequently affects how they vote. The authors suggest that the catalytic process works in much the same way that Trish Devine thinks people overcome the habit of prejudice, as we described in chapter 5. People sense a conflict between the values expressed

in the different decisions they make—they feel the dissonance—and are driven to find a way to resolve it. Putting panels on the roof resolves the conflict by signaling “I’m the kind of person who supports renewable energy.” Positive spillover is more likely when it [helps people see themselves as good](#).

There are a lot of caveats, of course. Negative spillover is a live threat, and plenty of experiments predict positive spillover only to find zilch. A spillover effect is only so strong, and the second action might be costly enough to quash it.

One way to think about the findings so far is as a proof of concept: small acts can add up, but not only in the way that votes add up or plastic bottles in a landfill do. Instead, they add up by affecting other, more consequential choices. Buying the right thing can seem like a trivial form of activism, but if it leads people to vote differently or join a movement, then it’s not so trivial after all.

Scaling Up

We began this book talking about how corporate PR directed attention to energy-efficient lightbulbs, litter, and recycling in order to distract attention from more impactful solutions. You might be wondering why we’re now defending some of those very same, much maligned consumer choices. The reason is *not* because we think human beings are going to recycle their way out of the climate crisis. Rather, we think that even these seemingly trivial acts have the potential to lead to big—often surprising—payoffs. Reshaping the energy grid is more important than recycling, but recycling might help us reshape the energy grid. The next chapter is about all kinds of different ways that little things—all the allegedly irrelevant goings-on at the individual level—can combine and grow and cascade into revolutionary change.

Cascades, Loops, and Tipping Points

From Agents of Change to Agents of Transformative Change

Small acts can have large effects. Sometimes the effects are personal, influencing one's future choices, while other times they are social, affecting other people's choices. Sometimes effects combine, and combine again, and combine again, and suddenly small acts become big cascades that change the world. When this happens, slowly at first and then seemingly all at once, a tipping point's been crossed.

Tipping points are all over the place. Novel viruses and dance crazes initially appear in isolated locations and then are suddenly everywhere; electric bikes and mobile phones remain niche luxury goods, and then overnight everyone has one; rumors and misinformation simmer at the fringes of society until a social media post goes viral.

Tipping points in social change are difficult to see coming. Everything seems stably arranged just before the quantum leap. There are lessons here for would-be changemakers, if we look in a both/and way.

The Great Escape

Back in chapter 3, we described humanity's "Great Escape" from abject conditions and poor health over the past 200 or so years. In 1820, four out of every five people lived in poverty, almost none lived in a democracy, and the vast majority spent their whole lives without learning to read. This had been the status quo for most of the 300,000-ish years that human beings had been around. The rate of change between what happened in the past 200 years and the rest of our species' existence is jaw-dropping. For 99.95 percent of that time, we lived on average about thirty-five years. The typical human life doubled in length in the last 0.05 percent of our species' existence. This didn't change at a continuous pace; it was a radical shift.

Even the most breathtaking tipping points like the Great Escape have their [origins in countless smaller acts](#) and events. Consider, for a moment, the problem of milk. Life expectancy in the United States in 1880 was barely above the global average, [about forty years](#). The overall average lifespan was kept low by a gruesomely high infant mortality rate, which at the time was trending in the wrong direction. In the middle of the nineteenth century, 60 percent of all deaths in New York were children under five, which was *up* from 30 percent in 1815. Life expectancy overall had also gone backward, declining by thirteen years from 1800 to 1850. The main culprit at the time appeared to be industrialization and the rapid growth of crowded cities.

One effect of this growth was to push farms further away from urban centers. An effect of that, and one of the reasons big cities became dangerous, was that milk now had to travel much farther from a cow's udder to a kid's lips—sometimes upward of hundreds of miles. All milk at that time was raw, and it spoiled fast.

Milk had already been a cause for concern because of the "swill milk" scandals of the 1850s, when dairies started feeding their cows residual mash from distilleries to save money. A *New York Times* editorial said swill milk was a "[bluish, white compound](#) of true milk, pus and dirty water, which, on standing, deposits a yellowish, brown sediment that is manufactured in the stables attached to large distilleries by running the refuse distillery slops through the udders of dying cows and over the unwashed hands of milkers." Some dairies were caught faking the look of cream at the top of milk bottles by adding a layer of pureed calf brains. Apparently the illusion was pretty convincing, right up until you poured the brains into a cup of coffee.

Both swill and ordinary milk were brimming with microorganisms, exposing babies to tuberculosis, typhoid, scarlet fever, diphtheria, and other diseases. Older kids who drank it were vulnerable as well. Contaminated milk contributed to the fact that [about one in four American kids](#) didn't make it out of childhood in 1890. One reporter at the time alleged that "the deaths of two-thirds of the children in New York and Brooklyn could be distinctly traced to the use of impure milk."

On a common telling, the invention of pasteurization saved the day. The story usually goes like this: the great French chemist Louis Pasteur was helping local distilleries solve problems with souring wine in the 1850s, which led him to discover that airborne microorganisms caused fermentation and . . . yadda yadda yadda . . . Pasteur came up with the germ theory of disease. At the request of Napoleon III in 1863, Pasteur began looking for solutions to wine contamination and discovered that heating it to about 130 degrees killed the microorganisms. He [saved the French wine and beer industries](#) before applying the process to milk.

The impression created by this story is that Great Man Genius Inventor Pasteur came along, discovered germs and pasteurization, and thus saved countless babies. It's not totally wrong. The process *did* save thousands upon thousands of lives. [Infant mortality plunged to 7 percent](#) in New York City in 1900, for example, thanks in part to the city's early adoption of pasteurization. But notice the dates. [Pasteurization didn't become standard practice](#) in the United States until a full half-century after Pasteur discovered it. The first state-wide requirement to pasteurize milk wasn't passed (by Michigan) until 1947!

Why did it take so long? Why wasn't pasteurization accepted or required by law right after Pasteur's discovery? Unsurprisingly, there were a lot of social barriers. Dairy farmers didn't want to foot the bill to install pasteurizers. Some worried that "cooked milk" wouldn't sell, and they might not have been wrong. Many people preferred the taste of raw milk, and folk theories about "good bacteria" being killed by pasteurization were common. The "sour milk cure," for example, was the idea that spoiled milk would "overcome" the bacteria in your body that was making you sick. (Ideas like these never went away. The website "goop" still recommends cleanses of raw goat milk to eliminate "parasites" in your gut.) Even the American Pediatric Society was disastrously late to the game, warning parents in 1899 that feeding pasteurized milk to their kids could give them scurvy.

It was a messy, decades-long battle, and the fighting wasn't just over pasteurization. Milk was one front in a larger war about consumer safety and the growth of the regulatory state. In 1906, Upton Sinclair's *The Jungle* was published, a searing exposé of Chicago meatpacking factories. That same year, the country took a big step forward in food safety as the federal Pure Food and Drug Act was passed into law.

An incredible change in human existence—the doubling of the average lifespan—was not due to scientific discovery alone. It required all sorts of people to fight for it, from journalists like Sinclair to philanthropists like Nathan Straus, who opened a series of milk distribution depots for poor children in New York City. These remained open for decades, helping to pave the way for normalizing pasteurization. They served many further purposes too. During the economic panic of 1893, the depots gave coal away to people who couldn't afford it, and in 1914–1915, they [served 1,135,731 penny meals](#) to the unemployed.

There's no telling what fraction of credit Straus deserves for popularizing pasteurization. Countless other overlooked figures contributed as well. Take Alice Evans, who rose from a rural Pennsylvania farm to become the first woman ever elected president of the American Society for Microbiology, in 1927. Her pioneering work on the *Brucella abortus* bacteria helped build scientific consensus about the dangers of unprocessed milk.

The Personal and the Social: Interacting Origins of Tipping Points

In the early twenty-first century, extraordinary changes in what it's like to be a gay person rippled through the US. Through most of the previous century, gay Americans often stayed in the closet. Those who came out risked being openly disowned by their families, exiled from their religious communities, fired from their jobs without legal recourse, mocked in popular culture, and regularly threatened with violence.

These risks remain, but they have lessened. Both institutions and individuals have changed.

Institutionally, in psychiatry, same-sex attraction is no longer pathologized, nineteen states have banned conversion therapy, and hate crime laws have been expanded to include discrimination on the basis of gender, sexual orientation, and gender identity. The Department of Defense abandoned “Don’t Ask, Don’t Tell” in 2011, and it was not long before marriage equality became the law of the land with the Supreme Court’s *Obergefell* decision in 2015 and the federal Respect for Marriage Act of 2022. Meanwhile, more churches than ever before are [welcoming gay congregants](#) and allowing them to become ministers.

Alongside these changes in laws, policies, and institutions, people’s minds changed too. In 2001, just 35 percent of American adults favored same-sex marriage; by 2019, [that percentage rose to 61](#). Both the scale and the breadth of these changes in attitude have been remarkable. Although not completely even, the direction of change is the same for Democrats and Republicans; whites, Blacks, Asians, Hispanics, [and Native Americans](#); men and women; and millennials, Gen Xers, baby boomers, and the Silent Generation. There’s still much further to go, and progress has been full of fits and starts and setbacks, but the US has come surprisingly far, surprisingly fast.

The institutional and individual aspects of this change are interlinked. As gay people gained visibility in culture, politics, and media, more and more gay people came out of the closet. One result of this was that more and more Americans came to understand that they already knew and loved people who were gay. More and more people’s attitudes changed, and the market grew for positive representations of gay people in culture, politics, and media. More fictional characters in film and TV started coming out too.

This kind of self-reinforcing cycle is a positive feedback process. A change in the system—public representations of gay people—pushes other elements of the system to change—people’s attitudes—which in turn creates more change in that first element—public representation. These spiraling changes in systems, attitudes, and representations eventually rippled out beyond the struggle for LGBTQ justice, inspiring activists in other social movements. These included migrants without documentation and people with mental health disorders who initiated campaigns to [“come out” to the wider world](#) as well.

There’s no “first element” in any of this, just like there’s no point where a circle “begins.” We could have reversed how we described the elements in this example, with changes in attitudes causing changes in representations that in turn create more changes in attitudes. But that’s just a matter of how we decided to tell the story.

Feedback Loops Defined

Going forward, how can we identify potential cascades and tipping points and nudge social change along? It’s easy to see in hindsight that [Nathan Straus](#) and [Will & Grace](#) were pivotal, but examples like these aren’t easy to translate into guidance here and now. “[Make a fortune](#) and create a highly effective charity to promote pasteurized milk” or “write a hit TV show that empowers people to come out” is a bit like the advice to “buy low and sell high”: not wrong, just too abstract to be useful.

One step is returning to basics to understand the underlying processes—like feedback loops—that drive these kinds of cascades and tipping points. A feedback loop is a process where the output of one part of a system feeds back and becomes part of the input to another part. Take audio feedback. When someone speaks into a microphone, the sound runs through the system, which then outputs it, amplified, from a speaker. But what if the sound from the speaker gets picked up by the mic because they’re too close together? Then that louder sound entering the mic gets run through the system and ultimately becomes even more amplified when it’s pumped out of the speaker . . . and can’t you just hear that screeching pitch now?

Musicians and public speakers do everything they can to prevent these escalating audio feedback loops, though even seemingly uncontrollable spirals of noise can be harnessed—Jimi Hendrix was a pioneer and master at [using feedback intentionally](#). Positive feedback is usually what people have in mind when thinking about tipping points and social change. The Great Escape and viral pandemics are examples of positive feedback.

“Positive” here doesn’t mean “good.” A process counts as a positive feedback loop when the input

pushes it to continue in the same “direction” as the previous state. In the audio example, each cycle of sound through the system makes it louder, amplifying and amplifying until it reaches as high of a volume as the system can sustain.

Unchecked, positive feedback loops are intrinsically unstable. The previous equilibrium is disrupted, changes build upon themselves, and an earlier state of the system is left behind. The changes will continue until the feedback loop is disrupted by some external factor, with the screech growing louder and louder until the sound system reaches its capacity—or someone pulls the plug. Think population explosions, inflation, the proliferation of cancer cells, and economic boom-and-bust panics.

So too are some of the more [terrifying courses that climate change could take](#). One likely contributor to a positive feedback loop of planetary heat is the thawing of permafrost. Around 20 percent of the Northern Hemisphere is permafrost, where the “perma” indicates that it’s so cold the soil stays frozen for at least two years straight. Permafrost stores gargantuan quantities of carbon. When it melts—which it is doing at alarming rates now thanks to warming global temperatures—soil microbes in permafrost could turn all that long-buried carbon into CO₂ and methane (which traps 25 to 86 times more heat than CO₂, pound for pound). The release of these gases and microbes could then further accelerate the planetary rise in temperature, not to mention risk unleashing another pandemic or two.

The other kind of feedback is negative. Again, “negative” here doesn’t mean “bad.” Negative feedback loops still involve processes in which the outputs of one cycle become the inputs of the next. But here the new input is subtracted from the previous output. Rather than amplifying a change, negative feedback processes often work to stabilize a system.

Thermostats illustrate this point. When a change is detected—heat or cold seep in from outside—a thermostat takes that input and produces an output that will counteract it to keep the room’s temperature steady—activating the AC or turning on the heat. The level of water in toilets, cruise control in cars, flight control systems on airplanes, and the regulatory systems in the human body use negative feedback processes, too.

In contrast to the inherent instability of positive feedback, negative feedback keeps things at a steady state as outside influences push it to change. The system aims to maintain equilibrium, “absorbing” and adjusting to whatever the world throws at it.

Positive feedback is the more familiar image when it comes to social change, as activists often want to kickstart tipping points. But maintaining some equilibria and preventing the wrong kinds of spiraling positive feedback can be just as worthy goals. Certainly, this is the case when it comes to physical systems like the Earth’s climate. The more humanity can do to subtract from the escalating climate catastrophe, the better.

Preventing the worst excesses of positive feedback is no less crucial in some social systems. Stabilizing democracy and reproductive rights spring to mind. Political scientists and historians have been sounding the alarm that the institutions protecting practices like these are far more difficult to create than they are to destroy.

Perhaps nothing puts the power of negative feedback into sharper relief than the threat of nuclear holocaust. The Cold War spiraled through a terrifying positive feedback loop in the latter half of the twentieth century, as the US and the Soviet Union strove to outdo each other with [bigger and better ways to blow up the world](#). But some combination of restraining forces and dumb luck have prevented nuclear holocaust (so far). There’s a both/and lesson here too.

Some of these restraining forces were people. Vasily Arkhipov was a Soviet officer in October 1962 when his submarine was stationed near Cuba. The sub had gone days without hearing from Moscow when the US Navy discovered it and began releasing depth charges to draw it to the surface. The sub captain quickly concluded they were under attack and that World War III was underway. He needed two more officers to agree to authorize launching their nuclear torpedoes, but he only persuaded one. The other was Vasily Arkhipov. He talked the captain down, and the sub came up. Arkhipov’s stabilizing efforts may have literally [saved the world](#).

Some systems need to be destroyed, others reconfigured, and still others stabilized. The rapid change in attitudes toward gay people and marriage equality in the US show we have the power to create progressive, not just positive, feedback loops. Vasily Arkhipov's story shows that there are even rare occasions when individual people find themselves with the power to stabilize an entire system and prevent the worst of all possible disasters.

Learning from Loops

There are no surefire ways to hasten or smother cascades of social change, but there are lessons to learn. Here are a few.

We May Be Very Far from Transformational Change; Then Again, We May Be Very Close

Working to create big structural change can be a lot like navigating a foreign land without a map. Would-be changemakers don't have landmarks to indicate how close or far they are from achieving their goals. Without a clear point of reference, even small increments of improvement can seem inconsequential, making it dispiriting to "settle" for them. This likely contributes to what many people [experience as "climate anxiety."](#) The problem feels enormous, and the proposed solutions are mind-stretchingly complex. Even when big victories are won, it's hard to be sure of their real impact. In 2022, the US passed the Inflation Reduction Act (IRA), which, despite its name, is less about inflation than about climate. It's the most significant piece of climate legislation the US has ever passed, and yet from a global perspective, even this can look like a drop in the bucket.

On the other hand, being in the dark about how far we are from a tipping point implies that truly transformational change [may not be that far off](#). Many people dedicate their lives to making change, finding their efforts thwarted over and over until a hidden tipping point is crossed. Ignaz Semmelweis, the "savior of mothers," discovered in 1847 that women's odds of surviving childbirth dramatically increased if doctors washed their hands. His research was roundly ridiculed for decades and hadn't been accepted yet when he died in 1865. That was right around the time Louis Pasteur discovered germ theory and . . . yadda yadda yadda . . . now doctors wash their hands dozens of times every day and (almost) all the milk is pasteurized. We're often largely in the dark about how far away big changes are. For all we know, there might be one right around the corner.

Flapping Your Butterfly Wings: The Unpredictability of Social Phase Changes

June 17, 1972, seemed like just another Saturday night of making the rounds for security guard Frank Wills. Then he noticed some duct tape on a door and realized it was preventing the latch from locking. Wills ripped it off and resumed his rounds, only to find another piece of tape on the same lock when he circled back a half hour later. He hustled up the stairs to the lobby of the office building, the Watergate, and called the cops. Without realizing it, and just by doing his job well, Wills had discovered agents of the Nixon administration trying to break into and bug the Democratic National Committee office. Wills's phone call started a sequence of events that would lead to President Nixon's resignation [two years and two months later](#).

Minor actions can set off cascades that lead, in a surprisingly short time, to major structural outcomes. This reflects a general feature of complex systems. Causal effects in such systems don't always build on each other in a smooth or continuous way. Sometimes they build nonlinearly, allowing seemingly small events to produce disproportionately large changes.

This is another reason that, as strong as it is, the temptation to compare the effects of individual actions to drops in a bucket can be profoundly misleading. When filling a bucket of water, each drop contributes the same amount. But interventions in complex systems don't work that way. Luckily, there's a better analogy nearby: when heating a bucket of water, not every degree of temperature change has the same

incremental effect as the one before it. Some are transformational. For example, freezing and boiling reflect chemical tipping points. When one of those is crossed, a phase transition happens, and the properties of the entire bucket of water shift quickly and dramatically.

Like water, the social world is a complex system rife with tipping points. However, unlike water, we rarely know where the tipping points in the social world are. As a result, there's no recipe for creating cascades of [change in complex social systems](#). But once again, there's another way to look at this. Like [the storied butterfly](#) that flapped its wings and caused a tornado halfway around the world, any of us could be on the brink of doing something that will change the world, even if we don't know where or when.

Being Conservative About Being Conservative About Social Change

Human beings have a long track record of earnestly trying to engineer incredibly complex systems to improve them, only to make things worse. The economist Horst Siebert gave a catchy name to these kinds of unintended consequences: [the cobra effect](#). The story goes that the colonial British government in Delhi was fed up with the proliferation of venomous cobras, so it offered bounties to hunters who brought in cobra skins. The bounty program seemed to be going great, right up until the Delhiites figured out that they could just breed the cobras themselves instead of hunt them. Once the Brits realized that the bounty had become a perverse incentive, they canceled the program, at which point people released the cobras that they had been breeding, making the population of wild cobras even larger than it had been before the bounty.

The cobra effect sharpens the bind we're in. There are big problems. We have to do *something* and probably something big. But the big things we have to do risk unintended consequences. Should people bioengineer mosquitoes so that they can't carry malaria? Should we research ways of spraying sulfur dioxide into the stratosphere to reflect sunlight [in hopes of cooling the overheated Earth](#)?

The philosopher and economist Friedrich Hayek spearheaded modern libertarian skepticism about human beings' ability to understand and control complex social systems. Hayek argued that people understand their social systems far less than they think. As a result, he recommended caution. Since no one has a comprehensive grasp of how large complex social systems work, if people enact reforms too recklessly or hubristically, they can make things worse rather than better.

We agree with Hayek in one sense. Aspiring changemakers can be [honest about what they don't know](#). This is especially true when pursuing structural change. One good way to grapple with that ignorance is by using the best tools available to investigate the likely effects of different proposals for change. That investigation should be informed, ideally, by empirical findings, modeling results, and detailed histories of relevant successes and failures. It's easy for all of us to breezily assume we know what works. But would-be changemakers should keep an open mind and be willing to adopt [an experimental mindset](#) that is sensitive to what data suggests about the downstream effects of their work. "Data" here can range from formal scientific studies to informal check-ins, with others and oneself, about whether a new policy seems to be helping, hurting, or doing nothing at all.

There is a duck to this rabbit, though, and a point where we think it's reasonable to get off the Hayek bandwagon. Yes, intellectual humility makes sense when it comes to intervening in complex systems. The effects of any specific effort to create change are difficult to predict. But so too are the effects of inaction, of not doing anything. There's no guarantee that our current course isn't heading into a cascade of looping effects that lead to disaster (and plenty of reasons to think it might be). We can't sit back and count on the world's negative feedback loops to keep things stable.

The consequences of action and inaction are both unknowable. This isn't to say they're the same. The status quo is more familiar than some hypothetical changed world. Elizabeth Kolbert's book *Under A White Sky* tells one tale after another of human beings falling prey to the cobra effect. A representative case: Central American cane toads were intentionally released in Australia in 1935 in hopes that they'd eat native beetles that were detrimental to sugarcane harvests. In the wild, the cane toads ignored the beetles but feasted on just about everything else. Today, there are about [200 million cane toads](#) in Australia, and

they're responsible for devastating populations of many native species, from northern quolls to goannas. (Quolls, goannas, and other animals get sick and die when they eat the toads, which are poisonous.) Before they released boatloads of imported cane toads into the outback, Australians had a pretty good sense of what a cane-toad-less future world would be like. (Pesty.) But they didn't have a good sense of what a future with them would be.

In the sorts of cases like racism and climate change that we've been concerned with in this book, expecting the future to resemble the past is a tougher sell. The status quo isn't static in these domains. There's no going back to how it was, but there's also no "new normal" on planet Earth as long as it keeps getting hotter and hotter, artificial intelligence grows smarter and smarter, and so on. *Not* taking action involves uncertainty and risk as well. What lies on the other side of that uncertainty and risk may be worse than the unexpected outcomes we risk by acting.

Incrementalism and Anti-Incrementalism

There's a reason coaches have to convince kids who are learning sports that the best they can do is improve little by little. The slow grind of incremental improvement is as unsexy as it gets. It's frustrating, unpleasant, and boring. Who doesn't wish they could have the program for Kung Fu instantaneously downloaded into their brain like Neo in *The Matrix*? (Alex gets as close as he can by listening to podcasts at 3.3 times speed.)

Change is hard to see. Your hair grows at a constant rate but mostly seems the same length from one day to the next. (Then one random morning—boom—you're in dire need of a haircut, Dan, you hippie.) In the same way, progress is difficult to see as progress when it happens incrementally.

One neat psychological study put this idea to the test. Across fourteen experiments and almost 11,000 participants, who shared their thoughts on twenty-five different issues, it found that people tend to have an all-or-nothing bias in their perception of change. They were asked, for example, to compare two manufacturing companies that had set out to become more sustainable. Neither company reached their sustainability goals after one year, but one had gotten much closer than the other. While some participants acknowledged that one company had done better than the other, many lumped both "failures" together as "all the same." Worse are the demotivating effects of this all-or-nothing bias. People tend to check out when they see all failures as the same. They become disenchanted and stop trying. They question the intent and commitment of those who didn't achieve "complete" change.

We need to find ways of countering this bias, and the stakes are high. If someone thinks surpassing 1.5°C of global warming is unavoidable—and as of 2025, it looks nearly inevitable they'd be right—they become more likely to think the game is up and there's no point in continuing to try to prevent further warming. They'd be disastrously wrong, though. Every fraction of a degree makes an enormous difference. But the salience of that clearly stated threshold has a psychological effect, making it seem like an all-or-nothing goal. It becomes all too easy to think that any efforts failing to limit warming to 1.5°C are futile, that they don't matter and all is lost, but also that if we succeed in limiting warming to 1.5°C, we can declare "Mission Accomplished" and the climate crisis solved; all the work is done. The first part is probably the most worrisome, though—once people set their sights on a goal, failing to meet it makes them [susceptible to giving up](#).

We suspect that this anti-incrementalism bias reverberates widely. Reducing infant mortality can feel like no progress at all as long as inequities in infant care related to race and poverty remain. Most people aren't aware that US carbon and greenhouse gas pollution has been [declining since its peak](#) between 2005 and 2007. Most people—across the political spectrum—think that Black incarceration rates have continued increasing since 2006, when they've actually declined by 35 percent.

These improvements are not nearly enough, but they're real and important. They're evidence that collective efforts to turn the tide can work. It's maddening when they go unreported and uncelebrated.

Incrementalism can be a real distraction, of course, like when fossil fuel companies—against all the best science and economics—try to convince us that we need to invest in natural gas as a "bridge fuel" before

renewables are ready for prime time. Advocating incrementalism in the face of massive injustice can invite the often-fair criticism that “justice delayed is justice denied.” But this shouldn’t cloud serious assessments of what counts as progress or be taken as reason not to celebrate small victories. The takeaway here is not about defending incremental progress. It’s about not getting defeated by it.

Onward to Organizing Back in chapter 1, we cited evidence that the film *An Inconvenient Truth* raised awareness about the climate crisis but failed to inspire palpable changes in the public’s behavior. (We speculated that its individualist recommendations for tackling the problem were partly to blame.) But the truth is we don’t really know for sure what incremental long-term effects the film has had on inspiring people to take action. It could have seeded ideas that took a while to bear fruit, whether or not social scientists ever figure out how to measure the film’s impact.

History has a way of defying expectations. In 2018, in the wake of yet another mass shooting, students in Parkland, Florida, walked out of their classes to demand gun reform. It might be tempting to see their efforts as just one more failure, and the effects their protests ultimately have on gun control will be difficult to say. But here’s one effect we know they did have: [inspiring a fifteen-year-old Swedish girl](#) by the name of Greta Thunberg to start a school strike to protest climate change.

One way unexpected consequences are brought about is by one social movement spilling over into another. That’s where we’ll head next, to take a both/and look at the promise and peril of social movements. If we’re going to light a fire for social change, it’s going to need little things like sparks and kindling and oxygen to get lit.